



Software Release Notes for the Acuitas Service Management System

Release 9.4.0.0

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Release Abstract

This document contains the software release notes for the Affirmed Networks Acuitas™ Service Management System, Release 9.4.0.0.

How to Obtain Product Documentation

Documentation is available for currently supported product releases. Documentation is available online in Adobe PDF format. You can view PDFs online using the Adobe Reader®. To download the latest version of the Adobe Reader software from the Adobe web site, go to <http://get.adobe.com/reader/>

You can access Affirmed Networks documentation from the SFTP server. Contact Affirmed Networks for instructions on how to access the SFTP server.

See the *Affirmed Networks Documentation Library* for a complete list of Affirmed Networks product documentation.

Technical Assistance Center

Affirmed Networks Technical Assistance Center (TAC) provides technical support 24 hours a day, 7 days a week (24 x 7) for any customer with an active maintenance contract.

To contact the TAC:



E-mail: prodsupport@affirmednetworks.com



24 x 7 Telephone support:

Local: 1-978-268-0871

Toll-free: 1-888-283-2838

Server Hardware

Minimum Hardware Requirements

The Acuitas server requires the following minimum configuration:

- HP DL380 G8 server
- Dual 1.2 Ghz Intel Sandy Bridge processors, ES-2658, 8-core, 70W
- 32 GB of DDR3, 1600 MHz
- *For new installations:* 4 x 300GB 6G SAS 15K RPM SFF (2.5-inch) drives
- 4 x Gigabit Ethernet copper RJ45 ports

For a list of the physical specifications of the Acuitas server hardware platform that ships from Affirmed Networks, see [Acuitas Server Specifications for Systems Shipped from Affirmed Networks](#).

Acuitas Server Specifications for Systems Shipped from Affirmed Networks

This section lists the physical specifications of the Acuitas server hardware platform that can be ordered from Affirmed Networks. Affirmed Networks supports the Acuitas server running on any of the following machines:

- HP ProLiant DL380 Gen9
- HP ProLiant DL360 Gen10 (with 64GB, 4 x 300GB drives)
- Blade of the HP BladeSystem c7000

Bare Metal Installation: If the server supplied by Affirmed Networks is for a bare metal installation, RAID 1+0 is used.

VM installation: If the server supplied by Affirmed Networks is for a VM installation, RAID 5 is used.

HP ProLiant DL380 Gen9

The following table lists the specifications for the SFF model of the HP ProLiant DL380 Gen9 server:

Table 1 HP ProLiant DL380 Gen9 Server Specifications

Specification	DL380 Gen9 Server
Physical	
Dimensions	■ Height: 3.44 in. (8.73 cm) ■ Width: 17.54 in (44.55 cm) ■ Depth: 26.75 in (67.94 cm)
Weight	Minimum: 32.6 lb (14.759 kg) Maximum: 51.5 lb (23.6 kg)
Power: Requirements per power supply	
	2 x 800 W power supplies
Voltage	Range line voltage: 100 to 120 VAC 200 to 240 VAC
Current	Power supply output: Rated Steady-State Power For 800W Power Supply: 800W (at 100 VAC), 800W (at 240 VAC)

Table 1 HP ProLiant DL380 Gen9 Server Specifications

Specification	DL380 Gen9 Server
Frequency	Maximum Peak Power For 800W Power Supply: 800W (at 100 to 127 VAC), 800W (at 200 to 240 1VAC), 800W (at 240 VAC)
Cooling	
BTUs per hour	3207 at 100V AC input 3071 at 200V AC input
Hardware	
Disk Size	4 x 300GB, SAS, 15K RPM, 2.5" HDD
CPU	2 x 12-core Intel Xeon Processor (E5-2680v3 2.5GHz 120W)
Memory	64GB (8 x 8 GB) Single Rank x4 DDR4-2133 CAS-15-15-15 Registered Memory
Ethernet Adapter	1GbE, 4P, 331FLR FIO Adapter
RAID	HP Smart Array P440ar/2GB/2GB FBWC controller
License	
iLO Advanced License	If you supply your own HP ProLiant DL380 Server (the system was not preconfigured from Affirmed Networks), you need to install the iLO Advanced license. To do so: 1 In a web browser, enter the iLO IP address. 2 Log into the iLO as user root. 3 Under Administration, click Licensing. 4 Enter the Advanced License Activation Key. 5 Click Install.

HP ProLiant DL360 Gen10

The following table lists the specifications for the SFF model of the HP ProLiant DL360 Gen10 server:

Table 2 HP ProLiant DL360 Gen10 Server Specifications

Specification	DL360 Gen10 Server
Physical	
Dimensions	Height: 1.69" (4.29cm)
	Width: 17.11" (43.46cm)
	Depth: 27.83" (70.7cm)
Weight	Minimum: 13.04 kg, 28.74 lb
	Maximum: 16.27 kg, 35.86 lb
Power: Requirements per power supply	
	2 x 800 W power supplies
Voltage	Range line voltage:
	100 to 120 VAC
	200 to 240 VAC
Current	Power supply output:
	Rated Steady-State Power
	For 1600W Power Supply: 1600W (at 240 VAC)
	For 800W Power Supply: 800W (at 100 VAC), 800W (at 240 VAC)
	For 500W Power Supply: 500W (at 100 VAC), 500W (at 240 VAC)
Frequency	Maximum Peak Power
	For 1600W Power Supply: 1600W (at 200 to 240 1VAC)
Cooling	
BTUs per hour	For 800W Power Supply: 3207 BTU/hr (at 100 VAC), 3071 BTU/hr (at 200 VAC)
	For 500W Power Supply: 1979 BTU/hr (at 100 VAC), 1911 BTU/hr (at 200 VAC),
Hardware	
Disk Size	4 x 300GB, SAS, 15K RPM, SC DS HDD
CPU	2 x 12-core Intel Xeon Processor (4116 Xeon-S 2.1GHz, 85W)
Memory	64GB (8 x 8 GB) Single Rank x8 PC4-2666V-R
Ethernet Adapter	10/25Gb 2-port 640FLR-SFP28 Adapter

Table 2 HP ProLiant DL360 Gen10 Server Specifications

Specification	DL360 Gen10 Server
RAID	HPE Smart Array P408i-a SR Gen10 controller
License	
iLO Advanced License	If you supply your own HP ProLiant DL360 Server (the system was not preconfigured from Affirmed Networks), you need to install the iLO Advanced license. To do so: 1 In a web browser, enter the iLO IP address. 2 Log into the iLO as user root. 3 Under Administration, click Licensing. 4 Enter the Advanced License Activation Key. 5 Click Install.

For additional information, including how to install the server using the commercial rackmount kit, see the appropriate model *HP ProLiant Server User Guide* available at the following web site: <http://www.hp.com/go/hpsc>

Minimum Acuitas Virtual Machine Specifications

This section lists the minimum specifications for Acuitas running as a virtual machine.

The following table lists the minimum virtual machine specifications.

Table 3 Minimum Virtual Machine Specifications

Specification	Description
Disk Provisioned Size	200GB <i>Note: If you have more storage available, you can increase the provisioned size once the Acuitas package has been deployed.</i>
CPU	16 vCPU <i>Note: If your system has the resources available, you can increase the number of CPUs once the Acuitas package has been deployed.</i>
Memory	32GB

Client Software Requirements

Acuitas supports the following client software:

Component	Software Description and Version
Operating System	■ Microsoft Windows 7
Browser	■ Latest version of Mozilla Firefox. Version 28.0 has been qualified. ■ Latest version of Google Chrome.
Other	■ Adobe Flash Player 10 and above (Firefox plug-in)

Server Software Requirements

Acuitas supports the following server software:

Component	Software Description and Version
Operating System	■ Red Hat Enterprise Linux, version 7.7 ■ Update to openssl-1.0.2k-12.el7 or later
Application Server	JBoss Wildfly 8.2.1 Is packaged with the Acuitas software package (ems-release.rpm)
Java	Redhat OpenJdk 8.0 (java-1.8.0-openjdk-1.8.0.222.b10-1.el7_7.x86_64) Is packaged with the Acuitas software package (ems-release.rpm)

Acuitas Capacity and Performance

This section lists the capacity and performance specifications of the Acuitas Service Management System.

Table 4 Acuitas Capacity and Performance

Capacity Metric	Specification
Number of Managed Network Elements	Based on the recommended server configuration listed in Table 1 , Acuitas can centrally manage up to 100 Network Elements, which can be a combination of MCCs and MCC MMEs.
Traps and Alarm Correlation	Acuitas can process traps and perform alarm correlation at the rate of 100 traps/sec sustained without any loss. This number is likely sufficient for handling traps from 100 or more Network Elements.
Performance Statistics	Acuitas collects bulk Performance Statistics at 5-minute intervals, which provides significant efficiencies over the traditional use of SNMP for such purposes. Collected statistics are stored in the database (up to 2 years) and aggregated into less granular statistics over time. This type of aggregation allows historical trending of performance statistics, while fixing the database size. Each Network Element requires approximately 100 MBs for storage of long term performance data, or approximately 10GB for 100 Network Elements.
Logs	Acuitas collects logs on demand for troubleshooting purposes and purges them automatically after one week, thus capping the disk space requirements.
Configuration Backups	Configuration Backup files are retained for a maximum of 90 days. The file size is typically 30MB total per device, or 3GB for 100 Network Elements.

Features in this Release

This release contains the following new features and enhancements.

See [Unsupported Features](#) for any restrictions on these features.

Feature	Description
Acuitas presentation of SSF config-role	SSF3.1.0.2 supports the new attribute /system/config-role with values slave/master/none. In the Network View tab, an expanded SSF device lists the /system/config-role node type of the device (MASTER, SLAVE, or displays nothing for NONE.) When you manage a Master or Slave SSF in the Configuration Navigator, the Configuration Navigaor label lists whether the SFF is a Master or Slave. When you attempt to save a configuration change on an SSF Slave, Acuitas now prompts you: Are you sure you want to modify configuration on an SSF slave node? To save the configuration, click OK. To cancel the change, click Cancel. In the Network View tab, when you attempt to delete an SSF Slave, Acuitas now prompts you: Are you sure you want to delete configuration on an SSF slave node? To delete the Slave, click OK. To cancel the deletion, click Cancel.

Feature	Description
Acuitas support to list vm-hostname instead of node-id in exported 5-minute PM CSV files See Include vm-hostname in Exported 5-minute PM CSV Files.	Acuitas support to list vm-hostname instead of node-id in exported 5-minute PM CSV files. To configure this setting, modify the PerformanceConfiguration.properties file in the following directory: <pre>/opt/Affirmed/NMS/server/ems/resources</pre> To configure this setting, use the following configurable attribute: <pre>DataCollection.insertdata.nodeinfo=[0 1]</pre> where: 0 sets node_id (Default) 1 sets vm_host_name Restart the Acuitas server for the new value to take effect. This setting affects only the PM tables with a column containing node-id information (Note: the column name varies by table): ■ CONTENTHTTPPROXYTCPOPTZSTATSSLOT ■ DISK_STATS ■ IPFRAGMODULEPERFSTATS ■ NATPOOLPERFSTATSSLOT ■ PORT_STATS ■ PROXYBUFFERSTATSSLOT ■ SLOT_STATS ■ SYSTEM_NODE_UPTIME ■ SYSTEMCPUSTATISTICS ■ SYSTEMCPUSTATISTICSCORELEVEL ■ SYSTEMCPUSTATISTICSCPULEVEL ■ SYSTEMMEMORYSTATISTICS ■ SYSTEMMEMORYSTATISTICSCPULEVEL ■ TASK_STATS ■ TCPPERFORMANCESTATSSLOT ■ TCPSPLICINGSTATSCONNECTIONSLOT When nodeinfo is set to 1, you can also specify the name used in the header column of the CSV file. To configure this setting, use the following configurable attribute: <pre>DataCollection.insertdata.nodeidHeader=vm_host_name</pre> where the default header is vm_host_name. The header column name can contain special characters hyphen (-) and underscore (_).
Acuitas upgrade to RedHat OpenJdk 8.0 See the <i>Acuitas System Administration Guide</i>.	Acuitas uses RedHat OpenJdk 8.0 (java-1.8.0-openjdk-1.8.0.222.b10-1.el7_7.x86_64) for EMS Java runtime
Requirement to reduce the number of audit log files	Acuitas reduced the number of kept audit log files from 100 to 50. This change reduces the maximum steady-state disk usage by audit logging below the existing threshold.

Feature	Description
Revised KPI formula for SGSN GPRS VM data throughput See Key Performance Indicators (KPIs) Metrics .	Revised the formula for the following KPIs: ■ SGSN - GPRS Uplink Throughput (Kbps) - Data VM: $((SD_RELAY.VS.SNDCP.InDataOctGb * 8) / samplePeriodInSeconds) / 1000$ ■ SGSN - GPRS Downlink Throughput (Kbps) - Data VM: $((SD_RELAY.VS.SNDCP.OutDataOctGb * 8) / samplePeriodInSeconds) / 1000$ ■ SGSN - GPRS Uplink Packet Throughput (Kpps) - Data VM: $((SD_RELAY.VS.SNDCP.InDataPktGb) / samplePeriodInSeconds) / 1000$ ■ SGSN - GPRS Downlink Packet Throughput (Kpps) - Data VM: $((SD_RELAY.VS.SNDCP.OutDataPktGb) / samplePeriodInSeconds) / 1000$
Accessibility Support Phase 2: Acuitas support for GUI refinements	Acuitas GUI refinements: ■ High contrast colors in charts: Added support to display chart elements of Dashboards and Reports in high contrast colors when user selects High Contrast Theme. In the tree, added tooltip to the tree node icons to display alarm severity. Added labels to the alarm filter icons on the toolbar to list severity. ■ Selectable Legends in charts: To improve visibility in displaying data points in charts (PM Dashboard, Historical/Realtime Reports, Detailed charts view, System Availability, Alarm Stats) added support to allow users to select/deselect instances in the legend to show/hide the respective data plots in a chart. ■ Different shapes and colors in line charts: To improve visibility and differentiate graph points in line charts, added different shapes and colors in charts. Legend instances and tooltips also display the respective shapes and colors. ■ Highlighting graph lines when hovering over legends: When hovering over legend instances, the corresponding graph plots are highlighted (marker size is increased). ■ Include frames in Dashboard slots: To improve visibility and differentiate Dashboard slots, added frames around Dashboard slots.
Add RADIUS peer statistics to the PM Dashboard See Key Performance Indicators (KPIs) Metrics .	In the PM Dashboard, added the following raw metrics to the RADIUSPROCEDURESTATSPEER metrics table. ■ Number of Account Request Starts Transmitted ■ Number of Account Request Interim Updates Tansmitted
Networks Configuration Backup support for MCC, MME, and SSF See Network Configuration Backup .	Added Acuitas support for backing up configuration data for each of the reachable managed NE types, including MCC, MME, and SSF: Tools => Network Configuration Backup The backup can be saved as plain text output and/or XML format output.

Feature	Description																																				
Acuitas Software Maintenance enhancements See Software Releases .	Enhancements to the Software Maintenance feature: Configuration => Software => Maintenance ■ For MME and SSF, Acuitas now lists a maximum of 10 software images for selection. For MCC, the maximum remains at 3. ■ In the Software Maintenance window, Acuitas added the Directory field to specify the path of the software files in the selected remote file server. Based on the selected element type (MCC/MME/SSF), remote file server, and path, Acuitas lists the appropriate software files for selection in the File field.																																				
Entitlement and SHIP KPIs See <i>Acuitas Performance Statistics</i> .	Acuitas support for the following new statistics table: SERVERPLUGINSTATSCHASSIS: ■ Server Plugin Summary Attempts ■ Server Plugin Summary Successes ■ Server Plugin Summary Failures ■ Server Plugin Detail HTTP Error Responses ■ Server Plugin Detail HTTP OK Responses ■ Server Plugin Failures Internal Failures ■ Server Plugin Failures Invalid Request ■ Server Plugin Failures No Consent Acuitas support for the following new KPI: ■ HTTP-Server - Success %: SERVERPLUGINSTATSCHASSIS.num_plugin_successes / SERVERPLUGINSTATSCHASSIS.num_plugin_attempts																																				
Acuitas supports the hyphen character (-) in Primary SMTP Host Name and Secondary SMTP Host Name in the Mail Server configuration See Configuring the Mail Server .	Configuring the Mail Server sets parameters that specify how the application sends messages to the mail server, including SMTP access and authentication settings. Acuitas now supports the hyphen character (-) in Primary SMTP Host Name and Secondary SMTP Host Name in the Mail Server configuration: Topology => Application Settings => Mail Server The host name fields can contain the special characters . and -.																																				
Acuitas support for management of MME13.1.4.3	Acuitas supports management of the Affirmed Networks MCC MME 13.1.4.3 including configuration commands and new registers in the following statistics group sets: <table><tr><th>Group Set</th><th>Group</th><th>Register ID</th><th>Register Name</th></tr><tr><td>s1</td><td>msg</td><td>84</td><td>S1.IncPwsRestartIndication</td></tr><tr><td>s1</td><td>msg</td><td>85</td><td>S1.IncPwsFailureIndication</td></tr><tr><td>s1Enb</td><td>msg</td><td>84</td><td>S1.IncPwsRestartIndication</td></tr><tr><td>s1Enb</td><td>msg</td><td>85</td><td>S1.IncPwsFailureIndication</td></tr><tr><td>sbc</td><td>pws</td><td>10</td><td>SBC.OutPwsRestartIndication</td></tr><tr><td>sbc</td><td>pws</td><td>11</td><td>SBC.OutPwsFailureIndication</td></tr><tr><td>sbc</td><td>sbc</td><td>14</td><td>SBC.OutWriteReplaceIndication</td></tr><tr><td>sbc</td><td>sbc</td><td>15</td><td>SBC.OutStopWarningIndication</td></tr></table>	Group Set	Group	Register ID	Register Name	s1	msg	84	S1.IncPwsRestartIndication	s1	msg	85	S1.IncPwsFailureIndication	s1Enb	msg	84	S1.IncPwsRestartIndication	s1Enb	msg	85	S1.IncPwsFailureIndication	sbc	pws	10	SBC.OutPwsRestartIndication	sbc	pws	11	SBC.OutPwsFailureIndication	sbc	sbc	14	SBC.OutWriteReplaceIndication	sbc	sbc	15	SBC.OutStopWarningIndication
Group Set	Group	Register ID	Register Name																																		
s1	msg	84	S1.IncPwsRestartIndication																																		
s1	msg	85	S1.IncPwsFailureIndication																																		
s1Enb	msg	84	S1.IncPwsRestartIndication																																		
s1Enb	msg	85	S1.IncPwsFailureIndication																																		
sbc	pws	10	SBC.OutPwsRestartIndication																																		
sbc	pws	11	SBC.OutPwsFailureIndication																																		
sbc	sbc	14	SBC.OutWriteReplaceIndication																																		
sbc	sbc	15	SBC.OutStopWarningIndication																																		

Feature	Description
Acuitas support for management of SSF3.1.3	Acuitas supports management of the Affirmed Networks SSF3.1.3 including configuration commands and the following changes. ■ Added support for the new attribute /system/config-role with values slave/master/none. ■ Added support for the following new counters: SSF_GTP_3GPP_V1_LEFT table: – create-pdpc-response-tx.reject.ssf.rest: The number of CREATE-PDP-CONTEXT-RESPONSE messages sent to the Gn-SGSN that are rejects initiated by the SSF due to reasons other than those explicitly defined in this statistics group. This is a breakout counter of the create-pdpc-response-tx.reject.ssf. – create-pdpc-response-tx.reject.ssf.rest.subscriberdb: The number of CREATE-PDP-CONTEXT-RESPONSE messages sent to the Gn-SGSN that are rejects initiated by the SSF due to REST based subscriber database query failure. This is a breakout counter of the create-pdpc-response-tx.reject.ssf.rest. SSF_GTP_3GPP_V2_LEFT table: – create-session-response-tx.reject.ssf.rest.sessiondb: The number of CREATE-SESSION-RESPONSE messages sent to the left-side nodes that are rejects initiated by the SSF caused by the REST based session database query failure. This is a breakout counter of the create-session-response-tx.reject.ssf.rest. – create-session-response-tx.reject.ssf.rest.subscriberdb: The number of CREATE-SESSION-RESPONSE messages sent to the left-side nodes that are rejects initiated by the SSF caused by the REST based subscriber database query failure. This is a breakout counter of the create-session-response-tx.reject.ssf.rest.
Acuitas support for binary backup format for MCC-MME devices	For backup, Acuitas supports only binary format for MCC-MME devices. For restore, Acuitas supports both xml and binary formats for MCC-MME devices.
Added support for the Sd interface between the PCRF and the GiGW (acting as TDF) See the <i>Affirmed Networks Gateway Administration Guide</i>.	The Sd interface is located between the PCRF and the Traffic Detection Function (TDF) and is used to detect traffic and report information on the detected application traffic. This feature adds support for the Sd interface between the PCRF and the GiGW (acting as a TDF) as defined in 3GPP 29.212, Release 15.3.0. The GiGW has a new Diameter session-learner interface used to establish and terminate TDF sessions to the PCRF.

Feature	Description
Added ability to provide variable QoS for traffic based on information retrieved from a manifest server and traffic detection using ADC over the Gx interface See the <i>Affirmed Networks Gx Interface Specification</i>.	This feature adds the ability to provide variable QoS for traffic based on information retrieved from a manifest server and traffic detection using ADC over the Gx interface. This feature adds support for the acwsponsorurl vendor-specific attribute (VSA) on the Gx interface. If the PCRF sends the acwsponsorurl VSA along with the Charging-Rule-Definition, TDF-Application-Identifier and Event-Trigger APPLICATION_START and APPLICATION_STOP in the CCA/RAR message, the MCC gateway will retrieve flow information from a manifest server and deliver this information to the PCRF via a CCR-U message. This feature adds a new parameter to configure a manifest interface profile. <pre>an3000-mcm-slot7cpu1(config)# service-construct manifest-interface-profile <manifest-interface-profile-name> admin-state <enabled/disabled> certificate-server ca-certificate-list <ca-certificate-list name> certificate-server certificate-url <cert server url/ip> certificate-server certificate-file-extension <1 to 255 char string> certificate-server network-context <network-context> certificate-server loopback-ip <local IP loopback> certificate-server crl-url <crl server url/ip> manifest manifest-max-size <5 to 512000> manifest manifest-max-size-unit <bytes/kbytes> certificate-verification admin-state <enabled/disabled></pre> The new manifest interface profile must be associated with a Workflow Control Profile. <pre>an3000-mcm-slot7cpu1(config)# workflow control-profile <control-profile-name> manifest-interface-profile <manifest-interface-profile-name></pre> This feature adds support for processing the Custom-Mute VSA received in CCA/RAR messages on the Gx interface. The Custom-Mute VSA is used to limit APPLICATION_START/STOP notifications sent to the PCRF. This feature also adds support for sending the Sponsored-Connectivity-Data AVP in the CCR-U message on the Gx interface. If the manifest response comes with a validity time, the MCC includes the validity time in the CC-Time AVP within the sponsored connectivity data and starts the validity timer for the manifest refresh.
Support for Dell R740 running ESXi 6.5 U2 with Fortville 25G NICs	This release adds support for the Dell R740 running VMware ESXi version 6.5 U2 with Fortville 25G NICs on MCC SR-IOV systems.

Feature	Description
Added user account password security improvements. See the <i>Affirmed Networks System Administration Guide</i>.	Added user account password security improvements in this release. Use the following command to configure password policies: <pre>an3000-mcm-slot7cpu1# user-access password-policies</pre> Use the following command to display the current password policies by domain: <pre>an3000-mcm-slot7cpu1# show user-access password-policies default domain policy Numeric Complexity : 0 range 0 - 6 Uppercase Complexity : 1 range 0 - 6 Lowercase Complexity : 1 range 0 - 6 Non-alphanumeric Complexity : 1 range 0 - 6 Limit on sequential characters : 0 range 0 - 64 Limit on repeated characters : 0 range 0 - 64 Allow username in password : 1 range 0 - 1 [0=no 1=yes] Minimum Password Length : 8 range 0 - 32 Password Dictionary : 0 range 0 - 2 [0=no 1=default 2=custom] Previous Password History : 3 range 0 - 30 Lockout Failure Limit : 5 range 0 - 6 Lockout Duration : 10 range 0 - 60 [minutes] Minimum Password Age : 0 range 0 - 7 [day(s)] Maximum Password Age : 0 range 0 - 365 [day(s)] admin domain policy (unset values use defaults)</pre>
Allow Martian routes to be advertised by BGP	Martian addresses are addresses reserved by IANA for special use. The following addresses are reserved for private networks and treated as Martian routes. 10.0.0.0/8, 172.16.0.0/12, 192.168.0.0/16, 169.254.0.0/16 This feature allows BGP to advertise Martian routes created for UE Pools and loopbacks. You can use route maps to suppress Martian routes.

Feature	Description
Enhanced tethering detection logic using GQUIC user agent fingerprinting	Enhanced tethering detection logic using GQUIC user agent fingerprinting. Added the following parameter to enable running the tethering detection algorithm using GQUIC user agent fingerprinting: <pre>an3000-mcm-slot7cpu1(config)# data-path dpi tethering-detection-algorithm gquic-user-agent <enable disable></pre> By default, this parameter is disabled. Changes to this setting take effect upon next reboot. Added the following parameter to configure the weight percentage for the GQUIC user agent quality index: <pre>an3000-mcm-slot7cpu1(config)# data-path dpi tethering-quality-index-result-weight percentage gquic-user-agent <percentage></pre> Added the following statistics to indicate the number of tethered flows detected by GQUIC user agent fingerprinting: <pre>an3000-mcm-slot7cpu1# show subscriber pdn-session <session-id> an3000-mcm-slot7cpu1# show zone default deep-packet-inspection statistics grid-level tethering <id> an3000-mcm-slot7cpu1# show zone default gateway statistics [cluster-level grid-level node-level] call-performance apn <id> an3000-mcm-slot7cpu1# show zone default gateway statistics [cluster-level grid-level node-level] call-performance pgw <id> tethered-flows-detected-using-gquic-user-agent</pre> In the FLOW EDR, added the GQUIC user agent value to the tetheringAlgorithm field: tetheringAlgorithm <string> {ttl, userAgent, tcpipfingerprinting, gquic-user-agent}
Proxy is TTL transparent See the <i>Affirmed Networks Value Added Services Guide</i> .	Previously, for upstream traffic from the UE to the origin server, IP packets that <i>traversed</i> the proxy had their TTL values reset to 63 for IPv4, and their hop limit values reset to 254 for IPv6. However, IP packets that <i>bypassed</i> the proxy had their TTL and hop limit values correctly decremented by one. With this feature, traffic traversing the proxy also has the TTL/hop limit value decremented. Specifically, for any HTTP, HTTPS, and QUIC flows originating from the UE for IPv4 and IPv6, the TTL or hop limit value is decremented by one, respectively. This applies to transparent IP and non-transparent IP traffic traversing the proxy, and to traffic bypassing the proxy, regardless of whether the User Space IP stack or the Linux IP stack is used. Note: OSU – During an upgrade, traffic traversing VM nodes that have not been upgraded will retain the existing behavior.

Feature	Description
Support for IP address filters See the <i>Affirmed Networks Gateway Administration Guide</i>.	This feature adds support for the ipv4-filter-list object, which filters IPv4 packets by matching one or more ranges of IP addresses. Prior to this release, you had to use the packet-filter object to filter packets by IP address ranges. The ipv4-filter-list is more efficient than the packet-filter when you are filtering only on IP addresses and a service-rule would require hundreds of packet-filters. You can configure 10 ipv4-filter-lists. An ipv4-filter-list supports IPv4. IPv6 is <i>not</i> currently supported. To configure an ipv4-filter-list: <pre>an3000-mcm-slot7cpu1(config)# service-construct ipv4-filter-list <name> filter-type [network ue] <CIDR range></pre> where: ■ filter-type [network ue] — Specifies whether the filter is for a network initiated connection or for a user equipment initiated connection. ■ <CIDR range> — Specifies one or more CIDR ranges for the IP addresses, for example, 3.1.1.0/24. You can enter 4000 CIDR ranges for each ipv4-filter-list. To include the ipv4-filter-list in a service-rule: <pre>an3000-mcm-slot7cpu1(config)# service-construct service-rule <name> ip-filter-list <name></pre>
Support for service-rule logical OR for matching conditions See the <i>Affirmed Networks Gateway Administration Guide</i>.	This feature adds the new evaluation-mode parameter to specify how many of the configured classifier types must be matched for a service-rule to be considered a match: <pre>an3000-mcm-slot7cpu1(config)# service-construct service-rule <name> evaluation-mode [AND OR]</pre> where: ■ AND – Each type of the configured classifiers must be matched. ■ OR – At least one type of the configured classifiers must be matched. Prior to this release, only the AND behavior existed.

Feature	Description
ePDG support for sending HTTP POST request messages using a configurable URI in the HTTP server configuration	This feature adds support for a configurable URI field (uri-path) in the HTTP server configuration. The ePDG uses the HTTP protocol to communicate subscriber information (IMSI) to a provisioning server. When the ePDG receives a Diameter-EAP-Answer (DEA) response from the AAA server with the result code, 5450-DIAMETER_ERROR_USER_NO_NON_3GPP_SUBSCRIPTION, the ePDG sends an HTTP_POST request with the subscriber IMSI included in the URL to the provisioning server. Configure the URI path: <pre>an3000-mcm-slot7cpu1(config)# service-construct http-server <name> uri-path <string></pre> Example: <pre>an3000-mcm-slot7cpu1(config-http-server-HTTP_TEST)# uri-path /soa-infra/services/Custom2D/ CreateVOWIFIProvisioningEPDGReqABCSImpl/HTTPPOSTCreateVOW IFIProvisioningIMSI=\$(IMSI)&operationName=Request-Response</pre> where: ■ uri-path is the server URI path. The server-ip-address and server-port values are added to the uri-path to create the URL for HTTP POST. If the uri-path is not configured, the request message uses the default URI format /imsi/\$(IMSI). ■ <string> values include: IMSI, MSISDN, HTTPVERSION, UEIPADDR, ORIGINALAPN, IMEI, ULICGI, RATTYPER, GWIPADDR, PLMNID, CHARGINGCHAR, SUBSCRIBERID, for example: \$(MSISDN). Associate the http-server-list with the appropriate http-server: <pre>an3000-mcm-slot7cpu1(config)# service-construct http-server-list <name> http-server <name></pre> Associate the http-server-list with the ePDG: <pre>an3000-mcm-slot7cpu1(config)# zone default gateway epdg <name> provisioning-server-list <server-list-name></pre>
Enhancements for flow-idle-timeout-filter See the <i>Affirmed Networks Gateway Administration Guide</i> .	The flow-idle-timeout-filter now supports the following configuration options: source IP address range, destination IP address range, and IP type. These new options are in addition to the existing destination port range and IP protocol options. To configure the new options: <pre>an3000-mcm-slot7cpu1(config)# service-construct flow-idle-timeout-filter <name> ip-version [v4-only v4-or-v6 v6-only] network-ip-address-end <address> network-ip-address-start <address> network-ip-v6-address-end <address> network-ip-v6-address-start <address> ue-ip-address-end <address> ue-ip-address-start <address> ue-ip-v6-address-end <address> ue-ip-v6-address-start <address></pre>

Feature	Description
Added support for modifying QCI/ARP values of the EPS dedicated bearer if the PCRF uses the Charging-Rule-Install AVP to modify QCI/ARP values See the <i>Affirmed Networks Gx Interface Specification</i>.	Each unique QoS Class Identifiers/Allocation and Retention Priority (QCI/ARP) combination identifies a bearer on the PGW. This feature adds support for the following two scenarios if the PCRF uses the Charging-Rule-Install AVP to modify QCI/ARP values. ■ If there are other charging rules on the existing dedicated bearer, the PGW keeps the existing dedicated bearer but removes the particular charging rule for which QCI/ARP values have been modified. The PGW creates a new dedicated bearer with the new QCI/ARP values to carry the traffic for this charging rule. ■ If there is already an existing dedicated bearer with the same QCI/ARP values as those in the charging rules being modified, the PGW moves those modified charging rules to the existing bearer and deletes the old dedicated bearer if no charging rules are left on that bearer.

Feature	Description
Added support for throttling control plane GTP messages on the PGW and SGW when the system is in an overload state See the <i>Affirmed Networks Gateway Administration Guide</i>.	Added support for throttling the control plane GTP messages on the PGW and SGW when the system is in an overload state. Throttling is based on configurable ARP priority levels. Emergency call setups are not affected by this feature. To configure throttling, use the new overload-control max-arp-priority-level command: <pre>an3000-mcm-slot7cpu1(config)# zone default gateway profile gateway-common overload-control max-arp-priority-level minor <priority> overload-control max-arp-priority-level major <priority> overload-control max-arp-priority-level critical <priority></pre> where <priority> is a bearer's ARP priority level. When the system is in an overload state, the thresholds are used as follows: ■ Create session requests are rejected if the default bearer ARP priority level is greater than the max-arp-priority-level. ■ Create bearer requests and bearer resource command messages are rejected if the session does not have any bearers with an ARP priority level that is less than or equal to the max-arp-priority-level. Note: All new session setups are rejected when the system is in overload state and the MCD table sync to standby process is in progress. An overload condition can be any of the following. If more than one overload condition exists, the highest overload state is used for throttling. ■ PGW and SGW tasks exceed the CPU overload states: – minor — 80 percent – major — 88 percent – critical — 95 percent ■ The number of PGW and SGW create-session-requests exceeds the configured 1, 5, or 15 minute thresholds. These thresholds are configured with the following command: <pre>an3000-mcm-slot7cpu1(config)# zone default gateway threshold node-level call-performance pdngw create-session-requests-[1 5 15]-minute-average major-enter <int> minor-enter <int> critical-enter <int> admin-state enabled</pre> ■ The node-level memory consumption percentage exceeds the configured 1, 5, or 15 minute thresholds. These thresholds are configured with the following command: <pre>an3000-mcm-slot7cpu1(config)# zone default system threshold node-level memory-stats csm-avg-[1 5 15]min-memory-util minor-enter <int> major-enter <int> critical-enter <int> admin-state enabled</pre>

Feature	Description
Added support for throttling control plane GTP messages on the PGW and SGW when the system is in an overload state (continued)	The node-level CPU consumption percentage exceeds the configured 1, 5, or 15 minute thresholds. These thresholds are configured with the following command: <pre>an3000-mcm-slot7cpu1(config)# zone default system threshold node-level cpu-stats csm-avg-[1 5 15]min-cpu-util minor-enter <int> major-enter <int> critical-enter <int> admin-state enabled</pre> The following command displays the new overload statistics: <pre>an3000-mcm-slot7cpu1# show zone default gateway statistics cluster-level call-performance [pgw sgw] include overload</pre> <pre>num-create-session-rejects-overload num-create-bearer-rejects-overload</pre> The following alarms are generated when the system is in an overload state: <pre>anBladeOverload anThresholdCrossingAlarm</pre>
Updated DPI script to validate user-defined signatures and use image management techniques to load the packages dynamically See the <i>Affirmed Networks DPI Script User Guide</i>.	The DPI script enables a Network Operator to add or update classification support for applications and protocols without requiring an MCC software update or restart. The DPI script validates user-defined signatures and increases system performance by using image management techniques to load the packages dynamically. Configure the dpiscript parameters as follows: <pre>an3000-mcm-slot7cpu1(config)# data-path dpi dpiscript</pre> - script-classification – Set to enabled or disabled (default). In the disabled state, user-defined signatures are loaded on the system but the classification/detection phase does not consider these signatures. - http-classification – Set to enabled or disabled for dpiscript http support. This parameter takes effect only when script-classification is enabled. - https-classification – Set to enabled or disabled for dpiscript https support. This parameter takes effect only when script-classification is enabled. See the <i>Affirmed Networks DPI Script User Guide</i> for information on configuring and creating a DPI script and collecting and analyzing pcaps.
Support for configurable Tethering Quality Index (QI)	In previous MCC releases, the Quality Index that rates the confidence level of DPI-detected tethering used hard coded values. This feature supports configurable values. A connection is marked as tethered only if the reported Quality Index (Confidence Index) is greater than or equal to the detection algorithm QI value configured using the following command: <pre>data-path dpi tethering-quality-index value <QI> [default = 70]</pre>
Added TTL value to the dpiStringInfo subfield of the FlowRecord field of the Flow EDR	Added the TTL value to the dpiStringInfo subfield of the FlowRecord field of the Flow EDR.

Feature	Description
Added support to encrypt non-transient in-memory and inter-VM identities for Lawful Intercept (LI) targets	Added support to use AES-256 to encrypt non-transient in-memory and inter-VM identities for Lawful Intercept (LI) targets, including: ■ Identities: IMSI, IMEI, MSISDN ■ Lawful Intercept Identifier (LIID) ■ Correlation ID The LI Administrator (ADMF) configures the AES-256 encryption key using the Public Key Infrastructure (PKI) CLI: <pre>pki install type calea name target-key encrypt object <key-text></pre> <pre>pki remove type calea name target-key</pre> where <key-text> is always 32 bytes and can be specified in one of the following formats: ■ raw bytes – 32 characters ■ base64 – 44 characters ■ hex – 64 characters Issue the following command to verify that AES-256 encryption is enabled: <pre>show calea service-status</pre> The command displays the encryption statistics, including: <pre>target-encryption-type aes256</pre> which indicates that the encryption is enabled.
Added support to use AES-256 to encrypt all Lawful Intercept (LI) information passed using PFCP over Sxa and Sxbc interfaces	Added support to use AES-256 to encrypt all LI information passed using PFCP over Sxa and Sxbc interfaces. These interfaces apply only to CUPS configurations. Previous Sxa and Sxbc interface encryption supported only MD5. The LI Administrator (ADMF) enables encryption: <pre>calea pfcplawfulinterceptie adminstate enabled encryption [disabled md5 aes256]</pre> (Default: disabled) In addition: ■ To enhance security, configure a secret-key: <pre>calea pfcplawfulinterceptie secretkey <key-text></pre> ■ If the secret-key <i>is not</i> configured, but an LI Target Encryption is configured, the specified AES-256 key is used as the secret-key: <pre>pki install type calea name target-key encrypt object <key-text></pre> ■ If neither is configured, a default hard-coded key is used.

Feature	Description
Added support for populating the UWAN User Location Information field in local, GTPP and Gz/Rf PGW CDRs when the UE Local IP Address and the UE UDP Port IE are received from the ePDG See the <i>Affirmed Networks Charging Data Records Specification</i>.	When the UE Local IP Address and the UE UDP Port are received from the ePDG, the MCC populates the UWAN User Location Information field in local (ASN.1 and CSV), GTPP and Gz/Rf PGW CDRs. This CDR field is configurable in local ASN.1 and GTPP Interim and Stop data record templates. The default <code>presentFlag</code> setting is <code>no</code>, indicating the data field is not present in CDRs. Top-level CDR field <pre>an3000-mcm-slot7cpu1(config)# service-construct data-record-template pgw ofcs gtp [interim stop] <template-name> [uwanuserlocationinfo uwanuserlocationinfoudpsourceport] presentFlag [yes no conditional]</pre> Within the Change of Charging Condition Field (List of Traffic Data Volumes) and the Change of Service Condition field (List of Service Data) <pre>an3000-mcm-slot7cpu1(config)# service-construct data-record-template pgw ofcs gtp [interim stop] <template-name> [changeofcharconditionuwanuserlocationinfo changeofcharconditionuwanuserlocationinfoudpsourceport changeofserviceconditionuwanuserlocationinfo changeofserviceconditionuwanuserlocationinfoudpsourceport] presentFlag [yes no conditional]</pre> This CDR field is configurable in local CSV Start, Interim and Stop data record templates. The default <code>presentFlag</code> setting is <code>no</code>, indicating the data field is not present in CDRs. <pre>an3000-mcm-slot7cpu1(config)# service-construct data-record-template pgw charging-local [start interim stop event] <template-name> uwanuserlocationinfo presentFlag [yes no conditional]</pre> The UWAN-User-Location-Information AVP is configurable in ACR data record templates. The default <code>presentFlag</code> setting is <code>no</code>, indicating the field is not sent. <pre>an3000-mcm-slot7cpu1(config)# service-construct data-record-template pgw ofcs diameter [start interim stop event] <template-name> uwanuserlocationinfo presentFlag [yes no conditional]</pre>

Feature	Description
Support for Multi-Access Edge Computing local breakout for CUPS See the <i>Affirmed Networks Gateway Administration Guide</i> .	Added support for Multi-Access Edge Computing (MEC) local breakout on a standalone CUPS SGW. Local breakout allows traffic that only needs local applications to be serviced at the operator edge site. This reduces latency for local applications, and reduces back-haul requirements. Configure local breakout on a CUPS standalone SGW CPF: Enable local breakout in a user-plane-profile. <pre>an3000-mcm-slot7cpu1(config)# zone default gateway user-plane-profile <name> mec-local-breakout enable</pre> Configure a user-plane-selector type of subscriber-analyzer that specifies the user-plane-profile that enables local breakout. <pre>an3000-mcm-slot7cpu1(config)# workflow subscriber-analyzer <name> analyzer-type user-plane-selector priority <priority> action user-plane-profile <name> key <list of keys></pre> When a key in this subscriber-analyzer is matched, the data-profile from the matching workflow-profile type of subscriber-analyzer is sent to the UPF. Data packets matching the data-profile undergo a local breakout.
Enhancements for ip-address-allocation-mode See the <i>Affirmed Networks Gateway Administration Guide</i> .	The ip-address-allocation-mode now supports the new options ipv4v6-prefer-v4 and ipv4v6-prefer-v6. The ip-address-allocation-mode specifies how the type of allocated address (IPv4 or IPv6) is chosen when the UE requests IPV4IPV6. To configure the new options: <pre>an3000-mcm-slot7cpu1(config)# zone default gateway apn <name> ip-address-allocation-mode [ipv4 ipv6 ipv4v6 ipv4v6-prefer-v4 ipv4v6-prefer-v6]</pre> New options: ■ ipv4v6-prefer-v4 – Allocate an IPv4 address if it is available; otherwise allocate an IPv6 address if it is available. If neither type of address is available, the session fails. ■ ipv4v6-prefer-v6 – Allocate an IPv6 address if it is available; otherwise allocate an IPv4 address if it is available. If neither type of address is available, the session fails. Options that already existed: ■ ipv4 – Allocate an IPv4 address if one is available. If an IPv4 address is not available, the session fails. ■ ipv6 – Allocate an IPv6 address if one is available. If an IPv6 address is not available, the session fails. ■ ipv4v6 – If a dual address bearer flag (DAF) is received from the UE, allocate both an IPv4 and an IPv6 address. If a DAF is not received, allocate an IPv4 address if it is available; otherwise allocate an IPv6 address if it is available. If neither type of address is available, the session fails. Note: If the UE requests IPv4 or requests IPv6, that type of address is allocated if it is available, regardless of the ip-address-allocation-mode. If the requested type of IP address is not available, the session fails.

Feature	Description
Added support for applications and protocols supported by the DPI Signature in MCC R9.4 and later releases	Added support for applications and protocols supported by the DPI Signature in MCC R9.4 and later releases. For details, see <i>Affirmed Networks DPI Classification Release Notes for R8.4 and Above</i>.
Added support for the CGI-SAI change and ECGI change event-triggers and increased the maximum number of SDCs that the gateway maintains per CDR from 20 to 50 See the <i>Affirmed Networks Charging Data Records Specification</i> and the <i>Affirmed Networks Gy Interface specification</i>.	This feature adds two new event-triggers to the <code>event-trigger-template</code>, which is used to specify the events that trigger an interim update on offline/online charging interfaces and to trigger the closure of an interim SGW/PGW/GGSN CDR. <i>Applies to the Gy, Gz/Rf and Ga (GTPP), local (ASN.1) interfaces.</i> For all event-triggers in the <code>event-trigger-template</code>, the maximum service data container (SDC) limit per CDR closure event is increased from 20 to 50. <pre>an3000-mcm-slot7cpu1(config)# service-construct interface-trigger-template event-trigger-template <template-name> [cgi-sai-change-close-trigger ecgi-change-close-trigger]</pre> ■ cgi-sai-change-close-trigger – Specifies whether a change in the Cell Global Identification- Service Area Identifier (CGI-SAI) triggers an update. Value 0 disables the trigger, value 1 - 50 enables the trigger. In the case of offline interfaces, the value specifies the maximum number of CGI-SAI change events to trigger a CDR close. ■ ecgi-change-close-trigger – Specifies whether a change in the E-UTRAN Cell Global Identifier (ECGI) triggers an update. Value 0 disables the trigger, value 1 - 50 enables the trigger. In the case of offline interfaces, the value specifies the maximum number of ECGI change events to trigger a CDR close. This feature also increased the maximum number of SDCs that the gateway maintains per CDR at the bearer-level from 20 to 50 if interim-updates are enabled on the interface. Upon reaching this limit, the gateway transfers the SDCs in an interim CDR to the offline server. <pre>an3000-mcm-slot7cpu1(config)# service-construct interface offline-charging <interface-name> [pgw sgw ggsn wag epdg] num-service-container [1..50]</pre> Added the following change in charging conditions to the Change-Condition AVP within the Service-Information -> PS-Information AVP sent on the Gz/Rf interface. 14 CGI-SAI Change 16 ECGI Change Added the following change in charging conditions to the Change-Condition ASN.1 field within the Change of Service Condition field sent on the Ga interface. cGI-SAIChange(21) eCGIChange(29) Added the following change in charging conditions to the Change-Condition ASN.1 field within the Change of Charging Condition field sent on the Ga interface. cGI-SAIChange(6) eCGIChange(10)

Feature	Description
Added support for the CGI-SAI change and ECGI change event-triggers and increased the maximum number of SDCs that the gateway maintains per CDR from 20 to 50 See the <i>Affirmed Networks Charging Data Records Specification</i> and the <i>Affirmed Networks Gy Interface specification</i>. Continued	Added two new configuration parameters under the offline-charging interface: <pre>an3000-mcm-slot7cpu1(config)# service-construct interface offline-charging <interface-name> [pgw sgw ggsn wag epdg] num-charge-condition-change [0..50]</pre> ■ num-charge-condition-change – Specifies the maximum number of change in charging conditions (such as RAT change, QoS change, ULI change, etc.) that the gateway maintains per CDR. Upon reaching this limit, the gateway transfers the SDCs in an interim CDR to the offline server. The default is 0. <i>Note: A trigger is only counted towards the num-charge-condition-change if it is enabled under the event-trigger-template.</i> <pre>an3000-mcm-slot7cpu1(config)# service-construct interface offline-charging <interface-name> [pgw ggsn] qos-change-trigger [pcrf-initiated access-nw-initiated any]</pre> ■ qos-change-trigger – Specifies when to activate the qos-change-trigger. – pcrf-initiated – Activate the qos-change-trigger for a PCRF-initiated QoS change only. – access-nw-initiated – Activate the qos-change-trigger for an access-network/UE-initiated QoS change only. – any – (default) Activate the qos-change-trigger for any type of QoS change.
Added support for the CAMEL Charging Information CDR field and added 3GPP 32.298 version 10.0 as an option for formatting local ASN.1 and GTPP CDRs See the <i>Affirmed Networks Charging Data Record Specification</i>.	This feature adds support for the CAMEL Charging Information field in PGW/GGSN local ASN.1 and GTPP CDRs. The CAMEL Charging Information CDR field is configurable in local ASN.1 and GTPP Interim and Stop data record templates. The default presentFlag setting is no, indicating the data field is not present in CDRs. <i>Note: The CAMEL Charging Information CDR field is only applicable for 3G sessions.</i> <pre>an3000-mcm-slot7cpu1(config)# service-construct data-record-template [pgw ggsn] ofcs gtp [interim stop] <template-name> camelinfo presentFlag [yes no conditional]</pre> This feature also adds version 10 as an option when configuring the 3GPP 32.298 version used to format local ASN.1 and GTPP CDR fields in Interim and Stop data records. To format local ASN.1 CDRs: <pre>an3000-mcm-slot7cpu1(config)# service-construct interface offline-charging <interface-name> [sgw pgw ggsn] local asn1-cdr-version [10.0 9.0 8.4 6.0]</pre> <i>Note: This parameter is only visible when the encode-format is set to asn1 or both.</i> To format GTPP CDRs: <pre>an3000-mcm-slot7cpu1(config)# service-construct interface aaa-interface-group gtp node-admin <node-name> version [10.0 9.0 8.4]</pre> Limitation When the version parameter is set to 9.0 for local ASN.1/GTPP CDRs, the Low Priority Indicator field in local ASN.1/GTPP Interim and Stop data record templates should be set to no if it is not needed. When the version parameter is set to 10.0 for local ASN.1/GTPP CDRs, the Low Priority Indicator field in local ASN.1/GTPP Interim and Stop data record templates should be set to yes (default).

Feature	Description
Added enhancements to the existing global firewall function. See the <i>Affirmed Networks System Administration Guide</i>.	This feature adds the ability to provide syslog for the following scenarios: ■ Summary of count of connections dropped per session due to incoming connections exceeding the max-connections threshold configured in workflow data-profile. ■ Summary of count of packets dropped due to malformed IP/TCP header. ■ Summary of count of flows dropped due to flow rate exceeding flow-rate-limit threshold configured in security policy rule. ■ Summary of count of packets dropped due to packet rate exceeding the packet-rate-limit threshold configured in security policy rule. The following limitations apply to this feature: ■ Upon SSM crash, firewall records that were not sent to the MCM are lost. ■ Upon MCM crash, firewall records that were not sent to the syslog server are lost.
Added a new configuration option under the online-charging interface to specify either incremental or accumulative volume-based usage reporting to the OCS See the <i>Affirmed Networks Gy Interface Specification</i>.	This feature adds a new configuration option under the online-charging interface to specify whether the MCC reports the CC-Total-Octets, CC-Input-Octets, and CC-Output-Octets (within the Used-Service-Units AVP) either accumulatively or incrementally. <pre>an3000-mcm-slot7cpu1(config)# service-construct interface online-charging <interface-name> usage-reporting-mode [accumulative-mode incremental-mode]</pre> ■ accumulative-mode – The accumulated CC-Total-Octets, CC-Input-Octets, and CC-Output-Octets is reported to the OCS in the Used-Service-Units AVP when the subscriber requests quota or when the PGW sends a CCR-U to report the final usage for the rating group. The CC-Total-Octets is reset to 0 only when the Gy session is terminated or the subscriber is cleared. ■ incremental-mode – (default) The incremental CC-Total-Octets, CC-Input-Octets, and CC-Output-Octets is reported to the OCS in the Used-Service-Units AVP each time the subscriber requests quota. The CC-Total-Octets is reset to 0 every time the PGW reports usage to the OCS. Note: Changes to this parameter take effect immediately. If this parameter is changed mid-session, it applies to all new subscriber sessions.
Added support for a new configuration option to configure the validity-time on the Gy interface	This feature adds a new online-charging configuration option to configure a validity time that is used when the OCS does not send the Validity-Time AVP within the MSCC AVP. <pre>an3000-mcm-slot7cpu1(config)# service-construct interface online-charging <interface-name> attributes validity-time <value-in-seconds></pre>
Improve DPI processing	This feature reduces MCC latency and bursty traffic pattern for DPI processing in the case of either packet drop or out of order packets.

Feature	Description
Added support for processing the Gx-TMO-Redirect-Server AVP when received from the PCRF in CCA-I/U and RAR messages See the <i>Affirmed Networks Gx Interface Specification</i>.	This feature adds support for the Gx-TMO-Redirect-Server AVP when received in the Charging-Rule-Definition AVP from the PCRF in the CCA-Initial (CCA-I), CCA-Update (CCA-U), or RAR message. When the PGW receives the Gx-TMO-Redirect-Server AVP within the Charging-Rule-Definition AVP from the PCRF, the MCC will perform a redirection for dynamic or externally-activated predefined rules. The redirection is only valid for HTTP URLs. This feature applies to online, offline, and metering rating groups. The following subfields are supported under the Gx-TMO-Redirect-Server AVP: ■ Gx-TMO-Redirect-Address-Type ■ Gx-TMO-Redirect-Server-Address ■ Gx-TMO-Append-Original-URL – If enabled, the MCC appends the original URL to the redirect URL. ■ Gx-TMO-Append-MSISDN – If enabled, the MCC appends the MSISDN to the redirect URL. ■ Gx-TMO-Append-IMSI – If enabled, the MCC appends the IMSI to the redirect URL. ■ Gx-TMO-Append-IMEI – If enabled, the MCC appends the IMEI to the redirect URL. ■ Gx-TMO-Append-MSIP – If enabled, the MCC appends the Mobil Station (MS) IP address to the redirect URL. – If both IPv4 and IPv6 MS IP addresses are allocated, the MCC appends the active IP address to the redirect URL. – For IPv6 IP addresses, the MCC uses the preferred format. ■ Gx-TMO-Deactivate-By-Redirect – If enabled, the MCC performs a one-time redirection. After the one-time redirection, the PGW will deactivate the rule. Note: If both the Redirect-Information AVP and Gx-TMO Redirect-Server AVP are received, the MCC will use the Gx-TMO Redirect-Server AVP for redirection and will ignore the Redirect-Information AVP provided the Gx-TMO-Redirect-Server AVP is enabled in the CCA data record template. The Gx-TMO-Redirect-Server AVP is configurable in Gx CCA data record templates. The default <code>presentFlag</code> setting is <code>no</code>, indicating the field is not processed when received in the CCA response. <pre>an3000-mcm-slot7cpu1(config)# service-construct data-record-template pgw pcrf diameter cca <template-name> tmoredirectserver presentFlag [yes no]</pre> This feature also adds a new configuration parameter under the advice-of-charge object to specify whether to include the MS IP address in the redirect URL. <pre>an3000-mcm-slot7cpu1(config)# service-construct advice-of-charge <name> http include-msip [true false]</pre>

Feature	Description
Added shaping enhancement for configurable chunk sizes	This feature provides a configurable option to control the data chunk size of rate-limited packets for the HTTP, HTTPS, and UDP protocols. The data size is controlled by the time interval at which rate-limited packets are sent to the UE for a given bitrate. For example: If the configured bitrate is 1.5Mbps and the configured send-interval is 50ms (0.05sec), MCC sends the following in each interval: $\text{chunksize} = (1.5\text{Mb} / 8) * 0.05 = 9375 \text{ bytes}$ A new send-interval field is added to the abr-optimization-profile. For example: <pre>services multi-protocol-proxy abr-optimization-profile AOP admin-state enabled send-interval – Configure rate limiting send interval (in ms) (min – 40ms, max – 500ms; in multiples of 10 ms; default – 100ms)</pre> A new send-interval field is added to the traffic-control-profile. For example: <pre>services multi-protocol-proxy traffic-control-profile ACP rate-limiting send-interval – Configure rate limiting send interval (in ms) (min – 40ms, max – 500ms; in multiples of 10 ms; default – 100ms) of 10ms; default – 100ms)</pre>
Added a new configuration option under the gateway service to configure the Node ID value used in SGW/PGW/GGSN CDRs See the <i>Affirmed Networks Charging Data Records Specification</i>.	This feature adds a new configuration option used to configure the Node ID value under the gateway service, which will be used as the Node ID value in local (ASN.1 and CSV), GTPP and Gz/Rf SGW/PGW/GGSN CDRs. If this parameter is not configured, the gateway service name is used as the value for the Node ID field in SGW/PGW/GGSN CDRs. <pre>an3000-mcm-slot7cpu1(config)# zone default gateway [sgw pgw ggsn] <gateway-service-name> node-id <node-id-string></pre> Note: The <i>node-id</i> parameter under the gateway service should not be configured or changed mid-session. If it is, the node-id in subsequent CDRs will contain the newly configured value. There is an existing configuration parameter under the offline-charging interface that allows you to specify whether the Node ID value in CDRs reports the configured gateway service name along with the Service Type ID or just the configured gateway service name. <pre>an3000-mcm-slot7cpu1(config)# service-construct interface offline-charging <interface-name> [sgw pgw ggsn] unique-cdr-nodeid [enabled disabled]</pre>
SGX Downlink Data Notification message to MME always provides IMSI See the <i>Affirmed Networks Gateway Administration Guide</i>.	SGW will always include the IMSI value in the Downlink Data Notification message if the MME provided the IMSI in the Create Session Request.

Feature	Description
Added ability to override the charging-characteristics (CC) value received in the Create Session Request message for a subscriber based on the subscriber type See the <i>Affirmed Networks Gx Interface Specification</i>.	This feature adds the ability to override the charging-characteristics (CC) value received in the Create Session Request message for a subscriber based on the subscriber type. Set parameters using the following commands. 1 Configure a charging-characteristics profile with a configured CC value for each subscriber type. <pre>an3000-mcm-slot7cpu1(config)# service-construct charging-characteristics-profile <cc-profile-name> [home-subscriber-cc-value roaming-subscriber-cc-value visiting-subscriber-cc-value] <value></pre> 2 Associate this charging-characteristics profile with a Charging-Rule-Base-Name (CRBN) profile, APN and/or gateway service. <pre>an3000-mcm-slot7cpu1(config)# services crbn-profile <crbn-profile-name> charging-characteristics-profile <cc-profile-name> an3000-mcm-slot7cpu1(config)# zone default gateway apn <apn-name> charging-characteristics-profile <cc-profile-name> an3000-mcm-slot7cpu1(config)# zone default gateway pgw <pgw-service-name> charging-characteristics-profile <cc-profile-name></pre> If the Create Session Request message contains a CC value, the MCC will override this value based on the following order of precedence. The MCC also uses this order of precedence for selecting the CC value. For example, if none of the CRBN profiles have a charging-characteristics-profile associated with them, the MCC will check the 3GPP-Charging-Characteristics AVP received in the RADIUS Access-Accept message. If the RADIUS Access-Accept does not provide a CC value, the MCC will check the charging-characteristics-profile associated with the APN, etc. 1 The CC value will be selected based on the charging-characteristics-profile associated with a CRBN profile. If multiple CRBNs are received, the MCC selects the first CRBN in the CRBN list. – CRBN profile received from the PCRF – CRBN profile received from the RADIUS server – The default CRBN profile (associated with a data profile) 2 The 3GPP-Charging-Characteristics AVP received in the RADIUS Access-Accept message. 3 The charging-characteristics-profile associated with the APN. 4 The charging-characteristics-profile associated with the gateway service. 5 The CC value received in the Create Session Request/Create PDP Context Request message. 6 The default CC value configured for the Control Profile. Note: If the CC value is not configured in the charging-characteristics-profile, the MCC uses the CC value received in the Create Session Request message.

Feature	Description
Support for monitoring the different establishment states of TCP and UDP flows accepted by a subscriber firewall flow See the <i>Affirmed Networks Gateway Administration Guide</i>.	TCP and UDP flows accepted by a subscriber-firewall-flow can be monitored for different establishment states. Separate timers for each establishment state are used to remove flows and to drop unexpected TCP messages. The timers are used as follows: tcp-open-timeout — Specifies the timeout period between each message involved in the three-way TCP connection establishment. During this period, if the expected message is not received, the flow is removed. tcp-established-timeout — Specifies the timeout period for no flow activity after the three-way handshake is complete. If no flow activity is detected during this period, the flow is removed. tcp-close-timeout — Specifies the timeout period between each message involved in at the four-way TCP close operation, starting with receipt of the initial FIN from either direction. During this period, if the expected message is not received, the message is removed. udp-open-timeout — Specifies the timeout period between receiving the initial UDP packet and receiving the response UDP packet. If the response UDP packet is not received during the timeout period, the flow is removed. udp-established-timeout — Specifies the timeout period for no flow activity after the response UDP is received. If no flow activity is detected during this period, the flow is removed. In addition, a non-configurable internal timer removes a flow 30 seconds after receiving an RST. The timers can be configured in two ways: ■ Global — The same global tracking profile is used to control the timers for any subscriber-firewall-flow that enables global control. ■ Local — A subscriber-firewall-flow specifies a particular tracking profile to control timers. To globally configure monitoring of the flow establishment states: 1 Enable global control for the tracking profile selection. <pre>an3000-mcm-slot7cpu1(config)# services subscriber-firewall subscriber-firewall-flow <name> flow-state-tracking-profile-selection global</pre> 2 Configure a flow-timeout-filter. See the <i>Affirmed Networks Gateway Administration Guide</i>. 3 Assign the flow-timeout filter to the flow-idle-timeout-rule that defines the timeout values. <pre>an3000-mcm-slot7cpu1(config)# service-construct flow-idle-timeout-rule <name> flow-idle-timeout-filter <name></pre> 4 Configure the timeout values for the connection states in the flow-idle-timeout-rule. <pre>an3000-mcm-slot7cpu1(config)# service-construct flow-idle-timeout-rule <name> tcp-open-timeout <seconds> tcp-established-timeout <seconds> tcp-close-timeout <seconds> udp-open-timeout <seconds> udp-established-timeout <seconds></pre>

Feature	Description
Support for monitoring the different establishment states of TCP and UDP flows accepted by a subscriber firewall flow (continued)	5 Assign the flow-idle-timeout-rule to a flow-idle-timeout-profile. <pre>an3000-mcm-slot7cpu1(config)# flow-idle-timeout-profile <name> flow-idle-timeout-rule <name></pre> 6 Assign the flow-idle-timeout-profile to the data profile. <pre>an3000-mcm-slot7cpu1(config)# workflow data-profile <name> flow-idle-timeout-profile <name></pre> All subscriber firewall flows that enable global monitoring use this flow-idle-timeout-profile. To locally configure monitoring of the flow establishment states: 1 Enable local control for the tracking profile selection. <pre>an3000-mcm-slot7cpu1(config)# services subscriber-firewall subscriber-firewall-flow <name> flow-state-tracking-profile-selection local</pre> 2 Configure the timeout values for the connection states in a flow-state-tracking-profile. <pre>an3000-mcm-slot7cpu1(config)# flow-state-tracking-profile <name> tcp-open-timeout <seconds> tcp-established-timeout <seconds> tcp-close-timeout <seconds> udp-open-timeout <seconds> udp-established-timeout <seconds></pre> 3 Assign the flow-state-tracking-profile to the subscriber-firewall-flow. <pre>an3000-mcm-slot7cpu1(config)# services subscriber-firewall subscriber-firewall-flow <name> flow-state-tracking-profile <name></pre>

Feature	Description
Added support for clearing NAT statistics	This feature adds commands for clearing NAT statistics. The only statistics that will not be cleared are concurrent counters, which display NAT resources that are currently being used in the MCC. For example, concurrent counters include connection-count and resource-utilization-session. The following command clears NAT statistics for all the nodes: <pre>services subscriber-firewall nat cluster-level module-statistics <chassis-id> clear</pre> The following command clears NAT statistics for a particular node: <pre>services subscriber-firewall nat node-level module-statistics <chassis-id> <node-id> clear</pre> The following command clears NAT statistics for a statistical nat-pool on a particular node: <pre>services subscriber-firewall nat statistical nat-pool-statistics <node-id> <network-context> <nat-pool> clear</pre> The following command clears NAT statistics for a sub-pool of a statistical nat-pool on a particular node: <pre>services subscriber-firewall nat statistical sub-pool-statistics <node-id> <network-context> <nat-pool> <sub-pool> clear</pre> The following command clears NAT statistics for a deterministic nat-pool: <pre>services subscriber-firewall nat deterministic nat-pool-statistics <network-context> <nat-pool> clear</pre> The following command clears NAT statistics for a sub-pool of a deterministic nat-pool: <pre>services subscriber-firewall nat deterministic sub-pool-statistics <network-context> <nat-pool> <sub-pool> clear</pre> The following command clears NAT statistics for a guaranteed nat-pool: <pre>services subscriber-firewall nat guaranteed nat-pool-statistics <network-context> <nat-pool> clear</pre> The following command clears NAT statistics for a sub-pool of a guaranteed nat-pool: <pre>services subscriber-firewall nat guaranteed sub-pool-statistics <network-context> <nat-pool> <sub-pool> clear</pre>
Added support for a predefined service rule to be activated under a dynamic rating group and for a predefined service rule under a rating sub-group to be moved to a dynamic rating group based on PCRF messaging. See the <i>Affirmed Networks Gx Interface Specification</i>.	This feature allows for a predefined service rule to be activated under a dynamic rating group. For example, if the quota-id received in the Charging-Rule-Definition AVP is not pre-configured on the MCC, the MCC will dynamically allocate a rating group with the received quota-id. If the Charging-Rule-Definition AVP also contains the name of a predefined service rule, the MCC will activate the predefined service rule under the dynamic rating group. This feature also allows for a predefined service rule under a rating sub-group to be moved to a dynamic rating group based on PCRF messaging. Note: Ensure the dynamic rating group's quota ID is not configured on the MCC.

Feature	Description
Factor additional latency for distance for congestion detection	This feature adds a configuration option for Round-trip Time (RTT) offsets when RTT is used in the congestion detection's throughput formula. RTT offsets can be configured in data centers for both <i>nearby</i> and <i>far away</i> subscribers to avoid misidentifying <i>far away</i> subscribers as congested. This feature reports the length of time that the client connection remains in each congestion level in the HTDR/EDR and also reevaluates policies for flows where the congestion levels change. This feature adds the following new HTDR/EDR fields in tcpClientConnectionInfo: ■ CongestionLevelNoneTime ■ CongestionLevelMildTime ■ CongestionLevelModerateTime ■ CongestionLevelSevereTime To configure RTT offsets: <pre>an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy rtt-location-offset-profile RTT-PROFILE</pre> Possible completions: <pre>compromise-rtt-offset RTT location offset for subscribers without a known TAC location-offset RTT location offset for subscribers with a known TAC</pre> <pre>an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy rtt-location-offset-profile RTT-PROFILE location-offset RTT-LOCATION1</pre> <pre>an3000-mcm-slot7cpu1(config-location-offset-RTT-LOCATION1)# ?</pre> Possible completions: <pre>rtt-location-offset RTT location offset(ms) tac-high-byte high byte of TAC</pre> <pre>an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy instance USTCP tcp-measurement</pre> Possible completions: <pre>admin-state Enable/Disable TCP Measurement congestion-level Congestion Level for applying Optimizations based on Client Side TCP measurement rtt-location-offset-profile RTT Location Offset Profile for this instance sampling-interval Client Side TCP measurement Sampling Interval (1/ms)</pre>
[NTT-DoCoMo, NTT-Communications] ESXi6.7 Support for SAEGW	[NTT-DoCoMo, NTT-Communications] ESXi6.7 Support for SAEGW

Feature	Description
Added support for the VM name to the TFDR and HTDR Recording Entity field.	If the vm-name is configured for a cluster node, the vm-name is included in the Recording Entity field in TFDRs and HTDRs. If the vm-name is not configured for a cluster node, a generic string, such as node_personality-node_number, will be used for the vm-name in TFDRs and HTDRs. For example, ISM-5, CSM-1, ASM-2, etc. The HHE header has a hard-coded limit of 2048 bytes. If the header addition with the vm-name exceeds this limit, the actual value in the header will get truncated to fit in 2048 bytes.
Added the ISM VMNAME on which the session has landed	Use the instructions in the <i>Affirmed Networks System Administration Guide</i> to configure the vm-name. If the vm-name is not configured, MCC uses a generic name. For example: <pre>an3000-mcm-slot7cpu1(config)# cluster <id> node <id> vm-name</pre> This feature adds the ISM VMNAME in the header. For example: <pre>an3000-mcm-slot7cpu1(config)# service-construct http-header-enrichment name request add-header header-string name value</pre> Possible completions: <pre><String, may include variable such as IMSI, MSISDN, HTTPVERSION, UEIPADDR, ORIGINALAPN, IMEI, ULICGI, RATTYPER, GWIPADDR, PLMNID, CHARGINGCHAR, TIMESTAMP, SUBSCRIBERID, VMNAME, example: \$(MSISDN)></pre>
Added support for specifying how the IP type is chosen when PGW-originated messages are sent to the ePDG See the <i>Affirmed Networks Gateway Administration Guide</i>.	This release adds CLI parameters that let you specify how the IP type (IPv4 or IPv6) is chosen when the PGW originates messages that are sent to the ePDG. In previous releases, the PGW always used IPv6 when it originated messages that it sent to the ePDG. To configure how the IP type is chosen, use the following command: <pre>an3000-mcm-slot7cpu1(config)# zone default gateway pgw <gateway-name> gtpc-endpoint-params dual-stack-behavior [initial-ip-type-used-by-peer ipv4 ipv6]</pre> where: ■ initial-ip-type-used-by-peer – The PGW should use the IP type that was used in the Create Session Request from the ePDG. ■ ipv4 – The PGW should use IPv4. ■ ipv6 – The PGW should use IPv6.

Feature	Description
Added ability to select an online-charging interface using the CRBN profile and to configure the time-threshold and volume-threshold for the online-charging interface See the <i>Affirmed Networks Gy Interface Specification</i>.	This feature allows you to select an online-charging interface using the CRBN profile if that CRBN profile has an online-charging interface configured. <pre>an3000-mcm-slot7cpu1(config)# services crbn-profile <profile-name> online-interface <interface-name></pre> Note the following guidelines when using this feature: ■ The online-charging interface configured for the CRBN profile overrides the online-charging interface configured for the workflow control profile. ■ If the MCC gateway receives multiple CRBN profiles from the PCRF or RADIUS server, the MCC uses the first CRBN profile in the list to select the online-charging interface. This feature also adds two new online-charging configuration options to configure the time-threshold and volume-threshold. The MCC uses these configured values when the OCS does not send the Time-Quota-Threshold and Volume-Quota-Threshold AVPs within the MSCC AVP and these values are not configured under the rating group. <pre>an3000-mcm-slot7cpu1(config)# service-construct interface online-charging <interface-name> attributes time-threshold <value> Volume-threshold <value></pre> Note the following guidelines when using this feature: ■ The time-threshold and volume-threshold configured for the rating group take precedence over the time-threshold and volume-threshold configured for the online-charging interface. ■ If the GSU is received with result code 4012, the MCC does not apply the time-threshold and volume-threshold configured for the rating group or online-charging interface.
Increased maximum MTU and MRU to 9100 bytes on all IP interfaces See the <i>Affirmed Networks Gateway Administration Guide</i> and the <i>Affirmed Networks System Administration Guide</i>.	IP interfaces on the MCC can now support an MTU and MRU as large as 9100 bytes. Prior to this release, the maximum MTU/MRU value was 1800. To configure the MTU/MRU size, first set the maximum packet size of the cluster with the new max-packet-size command: <pre>an3000-mcm-slot7cpu1(config)# cluster <id> max-packet-size <bytes></pre> This value has a default of 2048, and a range of 2048 through 9216. This value must be at least as large as the MTU value you want to use for the IP interfaces, and you must reboot the cluster. Next, set the MTU for IP interfaces: <pre>an3000-mcm-slot7cpu1(config)# network-context <id> ip-interface mtu <bytes></pre> This value has a default of 1500, and a range of 576 through 9100.

Feature	Description
Statistics support for non-200 HTTP responses	Added non-200 HTTP response statistics to the show services multi-protocol-proxy statistics [cluster-level node-level] detail command: Client Response Status Information: Client HTTP Response Status Information (1xx) Client Response Status Success OK: Client HTTP Response Status OK (200) Client Response Status Success: Client HTTP Response Status Other (2xx) Client Response Status Redirection: Client HTTP Response Status Redirection (3xx) Client Response Status Client Error: Client HTTP Response Status Client Error (4xx) Client Response Status Server Error: Client HTTP Response Status Server Error (5xx) Acuitas support for these statistics in the following statistics tables: CONTENTHTTPPROXYDETAILSTATSCHASSIS CONTENTHTTPPROXYDETAILSTATSSLOT
Added statistics on NAT connection failures	Added performance statistic at the APN level that indicates the total number of connection failures due to max-port-chunks exceeded for all sessions associated with that APN. show zone default gateway statistics cluster-level call-performance nat-conn-failure-max-port-chunks-exceeded Acuitas support for this statistic in the following statistics tables: APNCALLPERFSTATSGRID APNCALLPERFSTATSCHASSIS APNCALLPERFSTATSSLOT Added performance statistic at the NAT Pool level to track the cumulative number of port chunk allocation failures due to maximum chunks allocation limit for all sessions using that NAT pool. show services subscriber-firewall nat performance-statistics nat-pool nat-conn-failure-max-port-chunks-exceeded Acuitas support for this statistics in the following statistics table: NATPOOLPERFSTATSSLOT Added performance statistic at the NAT Pool level to track the per-session high watermark for the number of port chunks that were allocated on a specific slot. show services subscriber-firewall nat statistical nat-pool-statistics per-session-port-chunks-high-watermark
Linux debian update	Linux debian updates
Support for documentation listing the maximum number of supported Mobile Content Cloud object instances, by major MCC release	Documentation is available listing the maximum number of supported Mobile Content Cloud object instances, by major MCC release.

Feature	Description
Added a new configuration option to allocate a default time quota for use with the Default Gy Quota feature	Previously, the Default Gy Quota feature could only be used with rating groups that had the measurement-method set to volume. With this feature, the Default Gy Quota feature can also be used with rating groups that have the measurement-method set to time or timeAndVolume. Use the following command to configure a default time quota for use with the Default Gy Quota feature: <pre>an3000-mcm-slot7cpu1(config)# service-construct interface online-charging <interface-name> ocs-down-default-time-quota <value></pre> ocs-down-default-time-quota – Specifies the default time threshold, in seconds, allocated during an OCS failure when the rating group measurement-method set to time or timeAndVolume. The default is 90 seconds.

Feature	Description
Added support for dynamic rating groups (quota-id not pre-configured on the MCC) and the provisioning of rating sub-groups	Added the ability to dynamically allocate a rating group when the quota-id received in the Charging-Rule-Definition AVP is not pre-configured on the MCC. When the PCRF dynamically allocates a rating group, the MCC uses the Online and Offline AVPs received in the Charging-Rule-Definition AVP to enable the corresponding charging method for the rating group/service rule. This feature also adds the ability to configure a list of separate rating sub-groups under the same rating group (quota-id), provided the charging-method for the rating group is set to subgroup. A rating sub-group can have any of the following charging methods: ■ Only online charging ■ Only offline charging ■ Both offline and online charging ■ No charging Use the following commands to configure a rating group with rating sub-groups. 1 Configure a rating sub-group. <pre>an3000-mcm-slot7cpu1(config)# service-construct rating-sub-group <sub-group-name></pre> The following parameters are configurable for a rating sub-group. offline-enabled – Specifies whether to enable/disable offline charging for the rating sub-group. The default is true. online-enabled – Specifies whether to enable/disable online charging for the rating sub-group. The default is true. offline-volume-threshold – Specifies the volume threshold, in kilobytes, after which a new service-rule level service data container is generated for rules associated with this rating sub-group. offline-time-threshold – Specifies the time threshold, in seconds, after which a new service-rule level service data container is generated for rules associated with this rating sub-group. service-rule – Specifies the service rule(s) to associate with this rating sub-group. A service rule can only be associated with one rating sub-group within the same rating group. A service rule can be associated with two rating sub-groups as long as the two rating sub-groups are not associated with the same rating group. The service rule must already exist. 2 Associate the rating sub-groups with a rating group. <pre>an3000-mcm-slot7cpu1(config)# services charging rating-group <rating-group> charging-method subgroup rating-sub-group <list-of-sub-groups></pre> If the charging-method for the rating group is set to subgroup, you cannot configure a service rule directly under the rating group. Service rules must be associated with a rating sub-group. Note: The Override Control feature and synchronizing Gz/Rf and Gy Charging Data does not apply to rating sub-groups.

Feature	Description
Added support for new TMO-Transparent Data attribute for the RADIUS Access Accept Message and for the Diameter AVP on the Gy interface, and added support for a new CDR field on the Ga interface	Added support for the following: ■ Transparent data AVP in CCR messages. This AVP is configurable in Gy CCR data record templates and by default is set to no, indicating the field is not sent. <pre>an3000-mcm-slot7cpu1(config)# service-construct data-record-template pgw ocs diameter [event interim start stop] <template-name> tmotransparentdata presentFlag [yes no]</pre> ■ Transparent data attribute in RADIUS Access Accept Message from the RADIUS Authentication Server. This attribute is configurable in RADIUS authorization accept data record templates and by default is set to no, indicating the field is not processed. <pre>an3000-mcm-slot7cpu1(config)# service-construct data-record-template pgw auth radius access-accept <template-name> tmo_transparent_data presentFlag [yes no]</pre> ■ Transparent data subfield in the RecordExtensions field of GTPP CDRs (sent via the Ga interface). This subfield, transparentVSA, is configurable in GTPP data record templates and by default is set to no, indicating the field is not present in CDRs. <pre>an3000-mcm-slot7cpu1(config)# service-construct data-record-template [pgw sgw] ofcs gtp [interim stop] <template-name> extensioninfo presentFlag [yes no]</pre> Three other subfields have also been added to support the transparentVSA subfield: extensionType, length, and serviceList. To view the transparent data string for a subscriber on a PGW or SGW: <pre>an3000-mcm-slot7cpu1# show subscriber pdn-session <id></pre>
Support for PGW selection of serving network MCC/MNC and TMO-Transparent Data	This feature lets the PGW select MCC/MNC (PLMN ID) values in a configured priority order by using the GTP message from the SGSN/SGW, a sig-info-table, or the RADIUS Access Accept Message from the RADIUS Authentication Server. This feature also lets you configure the selection of TMO-Transparent Data values on the PGW when the RADIUS Access-Accept Message does not return a transparent data value. The PGW selects a value from a sig-info-table with PLMN ID mappings or from a sig-info-table with SGSN/SGW subnet mappings. This is enabled at the APN level.

Feature	Description
Enhanced the provisioning of the default charging method	This feature enhances the provisioning of the default charging method for a session. When the Online/Offline AVPs are included in the CCR-Initial (CCR-I) message at the command level, these AVPs indicate the default charging method for the session based on the configuration at the PCEF. This feature also adds support for the Online and Offline AVPs at the RAR command-level on the Gx interface. Currently, the user-subscription-charging-type parameter configured under the control-profile object is used to advertise the default charging method (online/offline) supported on the MCC. With this feature, the default charging method can also be configured under the APN object and/or the crbn-profile object and used to advertise the session-wide default charging method in the Online/Offline AVPs in the CCR-I message at the command level. However, the default charging method provided by the PCRF in the CCA-Initial (CCA-I) at the command level takes precedence over any pre-configured default charging method on the MCC gateway (PCEF). After processing the CCA-I, the MCC will derive the session-wide default charging method using the following order of priority. 1 Online/Offline AVP provided by the PCRF in the CCA-I at the command level. 2 Online/Offline configuration under the first activated CRBN profile in the CCA-I. 3 Online/Offline configuration under the APN. 4 Online/Offline configuration under the Control Profile. The MCC also now supports a PCRF-initiated mid-session change to the session-wide default charging method by including the Online/Offline AVPs in the RAR message at the command level. ■ If the PCRF changes online charging from enabled to disabled mid-session, the MCC reports all current outstanding usage to the OCS and disables online charging for the entire session. When online-charging is changed from disabled to enabled, the MCC starts tracking usage from that moment and reports the usage to the OCS. ■ If the PCRF changes offline charging from enabled to disabled mid-session, the MCC reports all current outstanding usage in service-rule level service data containers and disables offline charging for the entire session. When offline-charging is changed from disabled to enabled, the MCC starts tracking usage from that moment and reports the usage in offline CDRs. Set parameters for this feature using the following commands. The default is true. <pre>an3000-mcm-slot7cpu1(config)# zone default gateway apn <apn-name> charging-methods offline-enabled [true false] online-enabled [true false] an3000-mcm-slot7cpu1(config)# services crbn-profile <crbn-profile-name> charging-methods offline-enabled [true false] online-enabled [true false]</pre>

Feature	Description
Added a new configuration parameter to control the charging method for a service rule based on both the session-wide default charging method and the service-rule default charging method	This feature adds a new configuration parameter under the gateway-common profile to control the charging method for a service rule based on both the session-wide default charging method and the service-rule default charging method. <pre>an3000-mcm-slot7cpu1(config)# zone default gateway profile gateway-common <name> use-session-wide-default-charging-state-as-gating [enable disabled]</pre> ■ enabled – A rule will be charged using the online/offline charging interface if the Online/Offline AVP is enabled at both the session level and the service-rule level. Otherwise, the rule is not charged. ■ disabled – (default) A rule will be charged using the online/offline charging interface if the Online/Offline AVP is enabled at the service-rule level. This feature also adds a new configuration parameter to specify whether multiple Service-Identifiers can be included within the Multiple-Services-Credit-Control AVP on the Gy interface. <pre>an3000-mcm-slot7cpu1(config)# service-construct interface online-charging <interface-name> avp-customization mscc-serviceid num-instances [multiple single]</pre> This feature also adds a new configuration parameter to specify a rating group template to use when a dynamic rating group is created because of a charging rule install from the PCRF. <pre>an3000-mcm-slot7cpu1(config)# workflow data-profile <data-profile-name> dynamic-rg-template <rating-group-name></pre> dynamic-rg-template - Specifies the name of a rating group to be associate with the data profile. The rating group is used as a template when a dynamic rating group is created because of a charging rule install from the PCRF. The associated rating group must be configured with charging-method set to subgroup.
Support for OSPFv2 adjacency establishment between MCC shared-subnet interfaces and a peer router	This feature allows MCC interfaces that share an IP subnet to establish OSPFv2 adjacency with one peer router that is in the same subnet and same OSPF area and is not an Affirmed Networks device. The peer router must have only one IP interface in the same subnet as the MCC shared-subnet interfaces (many-to-one connectivity). The priority for all OSPF interfaces should be set to zero to support this feature.

Feature	Description
Added Acuitas Performance Manager support for new fastpath statistics	Added Acuitas Performance Manager support for new fastpath statistics: IPFRAGMODULEPERFSTATS: In addition to the statistics supported on the node level, the following statistics are now supported on the cluster level: numFragIpv4PktRcvd: Total number of fragmented IPv4 packets received numIpv4FragAged: Total number of IPv4 fragment reassembly CB aged numReassembledIpv4Pkt: Total number of IPv4 fragment reassembly complete numRefragmentIpv4Pkt: Total number of IPv4 refragment attempts numFragIpv6PktRcvd: Total number of fragmented IPv6 packets received numIpv6FragAged: Total number of IPv6 fragment reassembly CB aged numReassembledIpv6Pkt: Total number of IPv6 fragment reassembly complete numRefragmentIpv6Pkt: Total number of IPv6 refragment attempts numGetBufferFail: Total number of getBuffer attempts failure TCPREORDERPERFSTATS: tcp-reorder-holes-filled: TCP reorder holes filled tcp-reorder-retransmissions : TCP reorder retransmissions tcp-reorder-retransmissions-bytes: TCP reorder retransmission bytes tcp-reorder-future-segments: TCP reorder future segments tcp-reorder-segment-flushed: TCP reorder segments flushed tcp-reorder-segment-discarded: TCP reorder segments discarded tcp-reorder-buffer-discarded: TCP reorder buffer discarded tcp-reorder-total-buffers-queued: TCP reorder buffers queued tcp-reorder-no-trnsprt-header: TCP reorder no transport header counts tcp-reorder-sequence-num-mismatch: TCP reorder sequence number mismatch WORKFLOWPERFSTATS: Classifications: Number of classifications processed ReClassifications: Number of reclassifications processed ServiceRulesEvaluated: Number of service rules evaluated ServiceRulesBypassed: Number of service rules bypassed ServiceEvaluations: Number of services evaluated PacketFilterEvaluations: Number of packet filters evaluated HrgEvaluations: Number of HTTP Rule Groups evaluated HrgBypasses: Number of HTTP Rule Groups bypassed PargEvaluations: Number of Protocol Application Rule Groups evaluated PargBypasses: Number of Protocol Application Rule Groups bypassed ConnectionsAttempted: Number of connections attempted ConnectionsAllowed: Number of connections allowed ConnectionDelete: Number of connections deleted

Feature	Description
Added Acuitas Performance Manager support for new fastpath statistics (continued)	TcpConnectionsAttempted: Number of TCP connections attempted TcpConnectionsAllowed: Number of TCP connections allowed TcpConnectionDelete: Number of TCP connections deleted TcpUplinkPackets: Number of TCP uplink packets TcpDownlinkPackets: Number of TCP downlink packets UdpConnectionsAttempted: Number of UDP connections attempted UdpConnectionsAllowed: Number of UDP connections allowed UdpConnectionDelete: Number of UDP connections deleted UdpUplinkPackets: Number of UDP uplink packets UdpDownlinkPackets: Number of UDP downlink packets HttpConnectionsCreated: Number of HTTP connections created HttpConnectionsDeleted: Number of HTTP connections deleted HttpUplinkPackets: Number of HTTP uplink packets HttpDownlinkPackets: Number of HTTP downlink packets HttpsConnectionsCreated: Number of HTTPS connections created HttpsConnectionsDeleted: Number of HTTPS connections deleted HttpsUplinkPackets: Number of HTTPS uplink packets HttpsDownlinkPackets: Number of HTTPS downlink packets
Support for mapping a virtual APN to multiple real APNs	This feature adds support for mapping a virtual APN to a list of real APNs, and evenly distributing sessions among the real APNs. The virtual APN and the real APNs can be in the same network context or in different network contexts. The real APNs can all be in the same network context or in different network contexts. The real APNs should have UE IP pools of the same size. Configure the mapping by using the new mapped-apn-name-list command: <pre>an3000-mcm-slot7cpu1(config)# workflow control-profile <name> mapped-apn-name-list <list></pre> where: <list> is a list of real APNs to map to. You can list up to 128 APNs. To support this feature, you must also have APN mapping control set to explicit: <pre>an3000-mcm-slot7cpu1(config)# workflow control-profile <name> apn-mapping-control explicit</pre>

Feature	Description
Support for reporting performance statistics for the individual APNs that are mapped to an APN	This feature adds support for reporting performance statistics for the individual APNs that are mapped to an APN. By default, this feature is not enabled. Prior to this release, APN performance statistics could only be reported for the mapped APN. Note: The following configuration maps an APN to an APN: <pre>an3000-mcm-slot7cpu1(config)# workflow control-profile <name> apn-mapping-control explicit mapped-apn-name <name></pre> Configure statistics reporting for the individual, original APNs by using the new apn-statistics-control command: <pre>an3000-mcm-slot7cpu1(config)# workflow control-profile <name> apn-statistics-control [original mapped]</pre> where: original — Specifies that statistics are reported for individual APNs. mapped — (default) Specifies that statistics are reported for the mapped APN. To view APN performance statistics, use the following command: <pre>an3000-mcm-slot7cpu1(config)# show zone default gateway statistics cluster-level call-performance apn <name></pre> If apn-statistics-control is set to original, detailed APN performance statistics are shown for the individual, original APN. APN statistics for the mapped APN will still show statistics for IP address allocation activities, which are performed by the mapped APN (for example, remote-ipv4-reserve-successes and static-ipv4-reserve-successes). If apn-statistics-control is set to mapped, detailed APN performance statistics are shown for the mapped APN. APN statistics for an individual, original APN will still show statistics for UE connection activities, which are performed before APN mapping takes place (for example, create-session-attempts, total-redirected-apn-subscriber-count, and create-session-accepts-apn-matching-type-exact-match).

Feature	Description
Added support for the TMO-Virtual-Gi-ID VSA on RADIUS, Gx, and Gy Interfaces	This feature adds support for the TMO-Virtual-Gi-ID vendor-specific attribute (VSA) on RADIUS, Gx, and Gy interfaces. This feature also adds two new configuration parameters under the APN object to specify whether to process the TMO-Virtual-Gi-ID VSA when received in the Access-Accept message from the RADIUS server. <pre>an3000-mcm-slot7cpu1(config)# zone default gateway apn <apn-name> vsa-virtual-gi-id-enabled [true false] vsa-virtual-gi-id <default-value></pre> ■ vsa-virtual-gi-id-enabled – true – If the MCC gateway receives the TMO-Virtual-Gi-ID VSA in the Access-Accept message, it sends the VSA in Accounting-Request messages (Start/Interim/Stop) on the RADIUS interface and in CCR-I messages on the Gx and Gy interfaces. – false – (default) If the MCC gateway receives the TMO-Virtual-Gi-ID VSA in the Access-Accept message, it does not send the VSA in Accounting-Request messages on the RADIUS interface or in CCR-I messages on the Gx and Gy interfaces. ■ vsa-virtual-gi-id – Specifies a default value for the TMO-Virtual-Gi-ID VSA. If the vsa-virtual-gi-id-enabled parameter is set to true but the MCC gateway does not receive the TMO-Virtual-Gi-ID VSA in the RADIUS Access-Accept message, the MCC gateway uses this default value to fill the TMO-Virtual-Gi-ID VSA sent in RADIUS Accounting-Request messages (Start/Interim/Stop) and in CCR-I messages on the Gx and Gy interfaces. This parameter is only visible in the CLI when the vsa-virtual-gi-id-enabled parameter is set to true. You can configure whether to include or exclude the TMO-Virtual-Gi-ID VSA in RADIUS Access-Accept and Accounting-Request (Start/Stop/Interim) messages and Gx/Gy CCR-I messages. <pre>an3000-mcm-slot7cpu1(config)# service-construct data-record-template pgw auth radius access-accept <template-name> tmo_virtual_gi_id presentFlag [conditional no yes] an3000-mcm-slot7cpu1(config)# service-construct data-record-template pdngw accounting radius [start interim stop] <template-name> tmo_virtual_gi_id presentFlag [conditional no yes] an3000-mcm-slot7cpu1(config)# service-construct data-record-template pgw [pcrf ocs] diameter start <template-name> tmovirtualgiid presentFlag [yes no conditional]</pre>

Feature	Description
Added MCC support for new Uplink and Downlink data traffic statistics on the SGW by PLMN type	Added MCC support for new Uplink and Downlink data traffic statistics on the SGW by PLMN type: ■ Home PLMN ■ SNR (Seamless National Roaming) PLMN ■ Visiting/Roaming PLMN show zone <name> gateway statistics cluster-level call-performance sgw <name> (Performance Manager statistics table SGWCALLPERFSTATSGRID) num-sgw-downlink-packets-home-plmn num-sgw-downlink-bytes-home-plmn num-sgw-uplink-packets-home-plmn num-sgw-uplink-bytes-home-plmn num-sgw-downlink-drop-packets-home-plmn num-sgw-downlink-drop-bytes-home-plmn num-sgw-uplink-drop-packets-home-plmn num-sgw-uplink-drop-bytes-home-plmn num-sgw-downlink-packets-snr-plmn num-sgw-downlink-bytes-snr-plmn num-sgw-uplink-packets-snr-plmn num-sgw-uplink-bytes-snr-plmn num-sgw-downlink-drop-packets-snr-plmn num-sgw-downlink-drop-bytes-snr-plmn num-sgw-uplink-drop-packets-snr-plmn num-sgw-uplink-drop-bytes-snr-plmn num-sgw-downlink-packets-other-roaming-plmn num-sgw-downlink-bytes-other-roaming-plmn num-sgw-uplink-packets-other-roaming-plmn num-sgw-uplink-bytes-other-roaming-plmn num-sgw-downlink-drop-packets-other-roaming-plmn num-sgw-downlink-drop-bytes-other-roaming-plmn num-sgw-uplink-drop-packets-other-roaming-plmn num-sgw-uplink-drop-bytes-other-roaming-plmn

Feature	Description
Support for fragmentation of tunneled IPv6 packets exceeding the MRU size at the Gn interfaces.	This feature adds support for fragmenting tunneled IPv6 packets that egress through Gn interfaces and exceed the MRU size. Prior to this release, MCC supported fragmentation of only IPv4 packets egressing the Gn interfaces. To enable or disable fragmentation of IPv6 packets that egress Gn interfaces (disabled is the default), use the new fragment-tunneled-ipv6 command in the Gn network context: <pre>an3000-mcm-slot7cpu1(config)# network-context <name> fragment-tunneled-ipv6 [true false]</pre>
This release adds support for Google QUIC version Q046	MCC supports Google QUIC versions Q043, Q044, and Q046.

Feature	Description
Added ability to select a new control profile and online/offline interface using the CRBN from the PCRF	This feature adds the ability to select a new control profile and online/offline interface using the Charging-Rule-Base-Name (CRBN) AVP received from the PCRF. When the MCC gateway receives a CCA-I message from the PCRF containing the CRBN AVP, the MCC gateway can use the CRBN as a subscriber analyzer key to select a new subscriber analyzer. In doing so, the subscriber sessions use the control profile defined for the new subscriber analyzer. When a new control profile is selected using this method, the new control profile can only be used to make the following online/offline interface changes: ■ Online – Switch from a control profile with no online interface to a control profile with an online interface. ■ Offline – Switch to a control profile with a different offline interface. Only the following parameters can differ between the old offline interface and the new offline interface. – data-volume-limit – interim-update-interval – cdr-close-time Set parameters for this feature using the following existing commands. 1 Ensure the <code>cbrn-mode</code> parameter for the <code>pcrf-charging</code> interface is set to <code>data-profile</code>. <pre>an3000-mcm-slot7cpu1(config)# service-construct interface pcrf-charging <interface-name> cbrn-mode data-profile</pre> 2 Enable the MCC gateway to use the CRBN AVP received from the PCRF as a key to select a new subscriber analyzer. <pre>an3000-mcm-slot7cpu1(config)# service-construct interface pcrf-charging <interface-name> allow-cbrn-dataprof-based-subscriber-analyzer-reselecti on enabled</pre> 3 Configure a subscriber analyzer with a list of CRBNs (locally configured data-profiles) as one of the key attributes and associate a control profile with this subscriber analyzer. The control profile must already exist. <pre>an3000-mcm-slot7cpu1(config)# workflow subscriber-analyzer <subscriber-analyzer-name> analyzer-type workflow-profile action control-profile <control-profile-name> key <key-name> cbrn-dataprof-name-list values <list-of-data-profiles> match-type [match-any match-none]</pre>
Security improvements on 9.3 MCC	Applied security updates to address vulnerabilities on 9.3 MCC.

Feature	Description
Added support for overriding the static configuration of the destination-host and destination-realm on the Gx interface	The Diameter node on the MCC gateway communicates with statically configured Diameter peers via the Gx interface. When configuring a Diameter peer, you specify the peer host name and realm, which are used in the Destination-Host and Destination-Realm AVPs in Diameter request messages by default. You can now override these default settings on the Gx interface by configuring the override-dest-hostname and override-destination-realm parameters. When these override parameters are configured, the MCC gateway sends the CCR-Initial (CCR-I) to the selected Diameter peer but includes the configured override destination host/realm in the Destination-Host and Destination-Realm AVPs. The destination host/realm override configuration only applies to the Destination-Host and Destination-Realm AVPs in CCR-I messages. In CCR-U/CCR-T/ASA/RAA messages, the MCC gateway uses the values from the Origin-Host/Origin-Realm AVPs received in the CCA-Initial (CCA-I) message to fill the Destination-Host/Destination-Realm AVPs. If the override hostname is configured and if the CCA-I response does not have a different Origin-Host, the MCC gateway continues using the override hostname. Use the following CLI command to override the static configuration for the Destination-Host AVP sent in CCR-I messages on the Gx interface: <pre>an3000-mcm-slot7cpu1(config)# service-construct interface aaa-interface-group diameter node-admin <node-name> peer <peer-host-name> override-dest-hostname <destination-host></pre> Use the following CLI command to override the static configuration for the Destination-Realm AVP sent in CCR-I messages on the Gx interface: <pre>an3000-mcm-slot7cpu1(config)# service-construct interface pcrf-charging <interface-name> override-destination-realm <destination-realm></pre>
Added new configuration parameters to control whether the Destination-Host and Destination-Realm AVPs are updated based on information received in the CCA-I/CCA-U/RAR messages	With this feature, you can control whether the Destination-Host/Destination-Realm AVPs are updated in CCR-U/CCR-T messages on the Gx/Gy interfaces based on information received in the CCA-I/CCA-U/RAR messages. The default is enabled for the CCA-I/CCA-U messages and disabled for the RAR message. <pre>an3000-mcm-slot7cpu1(config)# service-construct interface [pcrf-charging online-charging] <interface-name> allow-dest-host-realm-change [cca-i cca-u rar] [enabled disabled]</pre> ■ enabled – If the received Origin Host/Origin Realm in the CCA-I/CCA-U/RAR is different from the current destination host and realm, the MCC uses the Origin-Host/Origin-Realm AVPs received in the CCA-I/CCA-U/RAR message to fill the Destination-Host/Destination-Realm AVPs in subsequent CCR-U/CCR-T messages. ■ disabled – The MCC gateway continues to use the existing values to fill the Destination-Host/Destination-Realm AVPs in subsequent CCR-U/CCR-T messages and does not update the Destination-Host/Destination-Realm AVPs with the values received in the CCA-I/CCA-U/RAR messages. Note: The MCC gateway always sends the RAA message with the Destination Host and Realm values of the Origin Host and Realm received in the RAR message. Note: For the CCA-I/CCA-U, the MCC will only update the Destination-Host and Destination-Realm in CCR-U/CCR-T messages if the result code received in the CCA-I/CCA-U is DIAMETER_SUCCESS (2001).

Feature	Description
Added a new online charging configuration option to restrict the PGW from establishing the online-charging session and requesting quota with the OCS until after it receives the first service data flow packet for rating groups configured with <code>credit-authorization-event</code> set to <code>bearerSetup</code>	Typically, the online-charging session with the OCS is established as part of the PDN session setup. With this feature, you can restrict the PGW from establishing an online-charging session and requesting quota until after it receives the first service data flow (SDF) packet for rating groups configured with the <code>credit-authorization-event</code> parameter set to <code>bearerSetup</code>. <pre>an3000-mcm-slot7cpu1(config)# service-construct interface online-charging <interface-name> first-packet-credit-auth-rating-groups [firstsdf all]</pre> ■ firstsdf – (default) The credit authorization for rating groups with the <code>credit-authorization-event</code> set to <code>firstSdfPkt</code> will be deferred until the PGW receives the first SDF packet. ■ all – For all active online rating groups, the credit authorization will be deferred until the PGW receives the first SDF packet. When the PGW receives the first SDF packet on any active online rating group, the quota request (MSCC AVP) is included in the CCR-U for all rating groups with <code>credit-authorization-event</code> set to <code>bearerSetup</code> and for the rating group (<code>bearerSetup/firstSdfPkt</code>) that received the first packet. For rating-groups with the <code>credit-authorization-event</code> set to <code>firstSdfPkt</code>, the credit authorization is requested when the PGW receives the first SDF packet for the rating group. <i>Note: This parameter is only visible when <code>interface-compliance</code> is set to <code>3GPPVersion2</code>.</i>

Feature	Description
Added consent-id to Secure HTTP Identify Protocol (SHIP) services	This feature enables closer coordination between Content Providers (CP) and end users who have given permission for that coordination. For example, if a CP requests to share subscriber information, this feature will verify if the user has consented to share their information with a given CP. If consent was granted, the data is shared; if not, access to subscriber information is rejected. This feature receives data from the PCRF. A new string named 'ConsentID' is added to the PCRF. That string is relayed to the Proxy for processing. If this feature is invoked and the content provider's configured consent-id matches a VSA string in the metadata for that user, then MCC processes the request. Otherwise, MCC returns an error code. This feature adds a new content provider and content status to data records and also adds the SERVER PLUGIN FAILURES NO CONTENT statistic: <pre>an3000-mcm-slot7cpu1# show services http-server statistics cluster-level server-plugin</pre> This feature adds the following new show subscriber pdn-session parameter: <pre>an3000-mcm-slot7cpu1# show subscriber pdn-session 239616 Session VSA ConsentID <Consent ID String></pre> This feature adds the following new show subscriber proxy-session parameter: <pre>an3000-mcm-slot7cpu1# show subscriber proxy-session detail 239616 Session VSA - ConsentID <Consent ID String></pre> This feature enables all proxy metadata across MCC to store up to 1022 bytes of data. Configure the consent-id as follows: <pre>an3000-mcm-slot7cpu1(config)# services http-server ship content-provider <CP-name> consent-id Consent ID for the content provider - string</pre> This feature filters SHIP requests as follows: ■ If the configured content provider consent-id is '*' (wildcard) (default), then the SHIP request is processed, regardless of the user strings supplied. The SHIP request could still fail due to other errors. ■ If the configured content provider consent-id is supplied and matches a VSA string provided for the user (not case sensitive), then the SHIP request is processed. The SHIP request could still fail due to other errors. ■ If the configured content provider consent-id is supplied but does not match a VSA string supplied for the user, then the SHIP request fails with the failure return code (1004).

Feature	Description
Added consent-id to Secure HTTP Identify Protocol (SHIP) services (continued)	In the HTTP Transaction data records (HTDRs), added the following field to the HTTPTransactionRecord: SHIPInformation Data Record: string Content Provider / uint64 AppID / not used ConsentStatus Consent Status / ConsentStatus contains the following enumerations: 0 = Consent Not Available 1 = Consent Not Matched 2 = Consent Matched ShipStatus Ship Status / ShipStatus contains the following enumerations: 0 = SHIP success 1 = Content Provider Not Found 2 = Info Not Available 3 = Consent Validation Failed 4 = SHIP Internal Server Error
Increase the number of supported service-rules	Increased the number of supported service-rules from 4096 to 16000. Increased the dynamic-pcc-rule range from 0x80000000 to 0xFFFFFFFF.
Support for ALGs without NAT	This feature adds support for the following ALGs when NAT is not performed on the source IP address and port for sent IP packets and is not performed on the destination IP address and port of received IP packets. ■ FTP — Supported for IPv4 and IPv6 ■ RTSP — Supported for IPv4 ■ SIP — Supported for IPv4 and IPv6 ■ TFTP — Supported for IPv4 and IPv6 To apply the ALGs to IP packets when NAT is not performed, use the subscriber-firewall-flow command: <pre>an3000-mcm-slot7cpu1(config)# services subscriber-firewall subscriber-firewall-flow <name> action pass alg [ftp rtsp sip tftp]</pre> where: action pass specifies that NAT is not performed on the packets. alg [ftp rtsp sip tftp] specifies the ALGs that should be applied to the packets.

Feature	Description
Added support to allow the PCRF to install/update up to 64 TFT packet filters on the default bearer locally (in Fastpath)	The PCRF can install/update Traffic Flow Template (TFT) packet filters on the default bearer using dynamic PCC rules, so that only traffic matching the packet filters is allowed to pass through the default bearer. This feature allows the PCRF to install/update up to 64 TFT packet filters on the default bearer locally (in Fastpath). These packet filters are used by Fastpath to control traffic but are not sent over GTP to the MME. To use this feature, ensure the <code>install-default-bearer-packet-filters-on-ue</code> parameter under the <code>gateway-common</code> object is set to <code>disabled</code>.
Added the ability for the MCC to establish/continue a session when the PCRF returns at least one valid Charging-Rule-Base-Name AVP in the CCA-I/CCA-U/RAR	Previously, if the PCRF returned multiple Charging-Rule-Base-Name AVPs and only one was predefined on the MCC, the MCC would terminate the session. With this feature, the MCC will establish/continue a session as long as the PCRF returns at least one predefined Charging-Rule-Base-Name AVP in the CCA-I/CCA-U/RAR message. Note: Applicable only when the <code>crbn-mode</code> parameter is set to <code>data-profile</code>.

VNF Manager Support Versions

VNF Manager has been tested in the following environments:

- ESXi version 6.5
- Red Hat OSP Newton and Queens
- vCenter version 6.5
- vCloud Director version 8.10 (vCloud Director API version 5.1.0)

Deprecated Device Model Versions in Acuitas 9.2.1.x

Acuitas 9.2.1.x and greater does not support the following device versions:

MCC Versions	MCC MME Versions	SSF Versions
■ 6.x ■ 7.0.x ■ 7.3.x ■ 7.5.x	■ 8.x ■ 10.x	■ 1.3

If you have any of these device versions, it is recommended that you upgrade to Acuitas 9.2.0 to maintain support continuity. If you upgrade to Acuitas 9.2.1 or greater, you lose support for these deprecated device versions.

Performance Statistics in this Release

This release contains the following new performance statistics. For details, see *Acuitas Performance Statistics*.

Table 5 Performance Statistics in this Release

Statistic Table and Statistics
APNCALLPERFSTATS
NumCreateBearerRejectsOverload
NumCreateSessionRejectsOverload
NumManifestServerRequestsSent
NumManifestServerResponsesRcvd
NatConnectionFailureMaxPortChunkExceeded
TetheredFlowsDetectedUsingGquicUserAgent
CONTENTHTTTPROXYDETAILSTATS
Client Response Status - Information (1xx)
Client Response Status - Success OK (200)
Client Response Status - Success Other (2xx)
Client Response Status - Redirection (3xx)
Client Response Status - Client Error (4xx)
Client Response Status - Server Error (5xx)
DPITETHERINGGRID
Tethered Flows Detected Using GquicUserAgent
DPIWEBHTTPTRANSACTIONSGRID
serverGetResponses
serverGetServiceTimeAvg
serverGetServiceTimeMovingAvg
proxyGetResponses
proxyGetServiceTimeAvg
proxyGetServiceTimeMovingAvg
EPDGAPNCALLPERFSTATS
NumSessionFailureAaaNoImsiForTempIdentity
NumSessionFailureAaaSessionValidation

Table 5 Performance Statistics in this Release

Statistic Table and Statistics
EPDGCALLPERFSTATS
NumSessionFailureAaaNoImsiForTempIdentity
NumSessionFailureAaaSessionValidation
NumDecryptionFailedAlgorithmFailure
NumDecryptionFailedBase64DecodeFailed
NumDecryptionFailedBase64StringTooLong
NumDecryptionFailedCertificateNotFound
NumDecryptionFailedNoPrivateKey
NumEncryptedIdentityLengthInvalid
NumIdentityDecryptionFailed
NumIdentityDecryptionSuccess
NumMessageFlowInitiatedByUeUsingFastReauthId
NumMessageFlowInitiatedByUeUsingPseudonymId
GTPPROXYAPNCALLPERFSTATS
NumCreateBearerRejectsOverload
NumCreateSessionRejectsOverload
NumManifestServerRequestsSent
NumManifestServerResponsesRcvd
TetheredFlowsDetectedUsingGquicUserAgent
GTPPROXYCALLPERFSTATS
NumCreateBearerRejectsOverload
NumCreateSessionRejectsOverload
NumManifestServerRequestsSent
NumManifestServerResponsesRcvd
TetheredFlowsDetectedUsingGquicUserAgent
GWSLINTERFACESTATS
NumDecryptionFailedAlgorithmFailure
NumDecryptionFailedBase64DecodeFailed
NumDecryptionFailedBase64StringTooLong
NumDecryptionFailedCertificateNotFound
NumDecryptionFailedNoPrivateKey
NumEncryptedIdentityLengthInvalid
NumIdentityDecryptionFailed
NumIdentityDecryptionSuccess
NumMessageFlowInitiatedByUeUsingFastReauthId
NumMessageFlowInitiatedByUeUsingPseudonymId

Table 5 Performance Statistics in this Release

Statistic Table and Statistics
NATPOOLPERFSTATS NatConnFailureMaxPortChunksExceeded
PGWCALLPERFSTATS NumCreateBearerRejectsOverload NumCreateSessionRejectsOverload NumManifestServerRequestsSent NumManifestServerResponsesRcvd TetheredFlowsDetectedUsingGquicUserAgent
SERVERPLUGINSTATS Server Plugin Summary Attempts Server Plugin Summary Successes Server Plugin Summary Failures Server Plugin Detail HTTP Error Responses Server Plugin Detail HTTP OK Responses Server Plugin Failures Internal Failures Server Plugin Failures Invalid Request Server Plugin Failures No Consent
SGWAPNCALLPERFSTATS NumCreateBearerRejectsOverload NumCreateSessionRejectsOverload
SGWCALLPERFSTATS NumCreateBearerRejectsOverload NumCreateSessionRejectsOverload
SLOT_STATS Isolated CPU Utilization Normalized Max
TASK_STATS Task Utilization
TCPPERFORMANCESTATS num_active_connection_openings num_passive_connection_openings num_failed_connection_attempts num_resets_received_for_embryonic_syn_rcv_sockets
UPLANECALLPERFSTATS NumSessionReportLateResponseDiscarded NumSessionReportResponseFailNoSession

Table 5 Performance Statistics in this Release

Statistic Table and Statistics
WAGAPNCALLPERFSTATS
NumSessionFailureAaaNoImsiForTempIdentity
NumSessionFailureAaaSessionValidation
WAGCALLPERFSTATS
NumSessionFailureAaaNoImsiForTempIdentity
NumSessionFailureAaaSessionValidation

Acuitas 8.1.1 Support for Real-time Statistics Reports for Earlier MCCs

Acuitas 8.1.1 does not support real-time statistics reports for MCC 8.0 and earlier and 8.1 releases for the following tables:

- IPFRAGMODULEPERFSTATS
- SIPBASICGRID
- SIPADVANCEDGRID

Acuitas 8.1.1 does not support real-time statistics reports for MCC 7.2.6 and earlier releases for the following tables:

- PROXYBUFFERSTATSCHASSIS
- PROXYBUFFERSTATSSLOT

Acuitas 8.0 Changes to Support New Dynamic Statistics Model for DPI Statistics

DPI changes affect the following areas in Acuitas:

- Generated Performance Manager data CSV files
- Real Time and Historical statistics and charts/reports

Changes to Performance Manager CSV Files

CSV File Name	Acuitas version 6.x, 7.x	Acuitas version 8.0 or higher
EMSDPIPROTOCOLSGRID	File columns: ■ timeofcollection ■ resourceid ■ zone_name ■ protocols_grid ■ name ■ category ■ flow_count ■ pkt_count ■ byte_count ■ pkt_ul_count ■ pkt_dl_count ■ byte_ul_count ■ byte_dl_count	File columns: ■ timeofcollection ■ resourceid ■ zone_name ■ protocols_name ■ protocols_groupName ■ protocols_protocolName ■ flows ■ uplink-packets ■ uplink-bytes ■ downlink-packets ■ downlink-bytes Note: On an 8.0 or higher Acuitas, MCC of any version uses these new column names Mapping of columns from pre-8.0 to 8.0: name => protocols_name category => protocols_groupName flow_count => flows pkt_ul_count => uplink-packets pkt_dl_count => downlink-packets byte_ul_count => uplink-bytes byte_dl_count => downlink-bytes

CSV File Name	Acuitas version 6.x, 7.x	Acuitas version 8.0 or higher
EMSDPIAPPLICATIONSGRID	File columns: ■ timeofcollection ■ resourceid ■ zone_name ■ applications_grid ■ name ■ flow_count ■ pkt_count ■ byte_count ■ pkt_ul_count ■ pkt_dl_count ■ byte_ul_count ■ byte_dl_count	File columns: ■ timeofcollection ■ resourceid ■ zone_name ■ applications_name ■ applications_groupName ■ applications_protocolName ■ flows ■ uplink-packets ■ uplink-bytes ■ downlink-packets ■ downlink-bytes <i>Note: On an 8.0 or higher Acuitas, MCC of any version uses these new column names</i> Mapping of columns from pre-8.0 to 8.0: name => applications_name flow_count => flows pkt_ul_count => uplink-packets pkt_dl_count => downlink-packets byte_ul_count => uplink-bytes byte_dl_count => downlink-bytes

CSV File Name	Acuitas version 6.x, 7.x	Acuitas version 8.0 or higher
EMSDPIOOPERATINGSYSTEMS GRID	File columns: ■ timeofcollection ■ resourceid ■ zone_name ■ operatingSystems_grid ■ name ■ osNetworkUsagePercentage ■ osNetworkUsage	File columns: ■ timeofcollection ■ resourceid ■ zone_name ■ operating-systems_name ■ operating-system_apn ■ flows ■ uplink-packets ■ uplink-bytes ■ downlink-packets ■ downlink-bytes <i>Note: On an 8.0 or higher Acuitas, MCC of any version uses these new column names</i> Mapping of columns from pre-8.0 to 8.0: name => operating-systems_name flow_count => flows pkt_ul_count => uplink-packets pkt_dl_count => downlink-packets byte_ul_count => uplink-bytes byte_dl_count => downlink-bytes
DPISUBPROTOCOL CSV files	A CSV file is generated for each sub-protocol For example: ■ DPINETWORKMANAGEMENT SUBPROTOCOLSLDAPGRID ■ DPIMOBILESUBPROTOCOLS BLACKBERRYGRID ■ DPIIMSUBPROTOCOLSJABBER GRID ■ DPIFILETRANSFERSUB PROTOCOLSOSUPDATEGRID	No change in the existing behavior

CSV File Name	Acuitas version 6.x, 7.x	Acuitas version 8.0 or higher
EMSDPIAPPLICATIONATTRS GRID	New CSV file (not available in pre-8.0)	File columns: ■ timeofcollection ■ resourceid ■ zone_name ■ applicationsAttrs_name ■ applicationsAttrs_groupName ■ applications_applicationName ■ applications_attributeName ■ flows ■ uplink-packets ■ uplink-bytes ■ downlink-packets ■ downlink-bytes
DPIGENERAL (No change in the existing behavior)		

Real Time and Historical Statistics

Table Name	Acuitas version 6.x, 7.x MCC version 6.x, 7.x	Acuitas version 8.0 or higher MCC version 6.x or higher
EMSDPIPROTOCOLSGRID	Not available in the Performance Manager	Table appears in the tree, but no data is collected
EMSDPIAPPLICATIONSGRID	Not available in the Performance Manager	Table appears in the tree, but no data is collected
EMSDPIOPERATINGSYSTEMS GRID	Not available in the Performance Manager	Table appears in the tree, but no data is collected
EMSDPIPROTOCOLSATTRS GRID	Not available in the Performance Manager	Table appears in the tree, but no data is collected
EMSDPIAPPLICATIONATTRS GRID	Not available in the Performance Manager	Table appears in the tree, but no data is collected
DPISUBPROTOCOL CSV files	All the SUBPROTOCOL tables are displayed in the Performance Manager (one table for each protocol)	No change in the existing behavior
DPIGENERAL (No change in the existing behavior)		

Unsupported Features

The following features are either unqualified or unsupported in this release. The features may appear in the documentation and in Acuitas and the CLI, but are not fully supported in this release:

Unsupported Acuitas Features

- IPv6 is not qualified in this release.
- Auto-VNF-Instantiate Commissioning Tool is no longer supported. It has been replaced by the Acuitas Virtual Network Function (VNF) Manager, which performs full life-cycle management of MCC clusters.
- NE Backup and Restore (right-click an NE and select **NE Backup and Restore**) is no longer supported. It has been replaced by NE Configuration Backup and Restore (**Tools => NE Configuration Backup**).
- Enhanced subscriber trace
- Enhanced **ping** and **traceroute**
- Dynamic scale out/scale in of a cluster appears in the Acuitas VNF Manager but is not yet supported
- VNF Manager does not support (on OpenStack in MCC MME clusters) deployments for LB/DATA/EDP nodes/VMs with or without SR-IOV enabled
- VNF Manager does not support deployment of MCC cluster without E/W redundancy if L3 E/W and Base network are enabled
- VNF Manager on OpenStack support of MCC clusters with all networks supporting SR-IOV has not been qualified.
- Acuitas management of MME 8.x and 9.x is no longer supported
- In the Protocol Application Rule Group, only the flow attribute values of audio and video are qualified.
- The Video Adaptation service is not supported in versions 8.2 and greater.
- Acuitas support for HPE HSS traps and alarms is qualified using a trap simulator, not an HSS device.

Unsupported General MCC Features

- In clusters with ISMs, user-space-networking is always enabled. Kernel-based networking is not supported.
- OSU from releases prior to 9.2.2.2.
Note: OSU from 9.2.2.2/9.3.0.2 and later to 9.4 is supported.
- ePDG and TWAG is not supported in the 9.3 and 9.4 releases.
- Geo-redundancy with pre-9.2 releases is not supported. Upgrade of pre-9.2 geo-redundant clusters will lose sessions.
- content-cache with ISM and user-space-networking
- Data profile switching (for example, triggered by MVNO, CC group, RAT change, etc.) along with CRBN profile functionality.
- CRN and CRBN cannot activate the same service rule.
- Scale In/Scale Out is not supported for CUPS and non-CUPS for all VM personalities.
Note: Manual Scale Out is supported only for ASMs/CPMs/SSMs in a CUPS cluster.
 - The **disabled** and **maintenance** options for **cluster <id> node <id> admin-state** are not supported. See [Update to Configuration Options for Cluster > Node > Admin State](#).
- In the **protocol-application-rule-group**, only the flow attribute values of audio and video are qualified.
- Support for the **apn-reporting** parameter under the RADIUS authorization and accounting interfaces has not been qualified.
- 5G NSA: Modify Access Bearer Request messages.
- IPv6 BFD and BGP peering is not supported for single-hop shared subnets.
Note: IPv4 BGP and BFD peering is supported for single-hop shared subnets.
- ISM back-end forwarding (BEF).
- SR-IOV base devices when management devices are non-SR-IOV is not qualified for RedHat OpenStack or VMware.
Note: SR-IOV base devices when management devices are non-SR-IOV is supported for Ubuntu OpenStack Newton.
- VoLTE SIP statistics are not supported on standalone SGW systems:
`an3000-mcm-slot7cpu1# show data-path statistics sip-statistics`

- MCC does not support geo-redundancy for GiGW/WAG sessions and GTPv0 sessions.

Unsupported CUPS Features

- IPv6 Prefix Delegation.
- UPF geo-redundancy.
Note: CPF geo-redundancy is supported.
- GTP-U F-TEID allocation on CPF.
- Rf-Gy sync is not supported for CUPS.
- 2G GTP v0 calls are not supported on CUPS; specifically, GTP v0 sessions on the UPF are not supported.
Note: 2G GTP v1 calls are supported on CUPS/UPF. Also, 2G GTP v0 calls can be terminated on the CPF as non-CUPS calls if GTP-U interface(s) are available on the CPF.
- UPF selection based on DNS query.
- UPF selection based on dynamic UPF load.
- Dual stack: IP v4/v6 is not supported on the PFCP (Sx) interface.
- Standalone Sxc, that is, standalone TDF-C and TDF-U.
- Co-located SGW-U and PGW-U sessions when selected by different/independent SGW-C and PGW-C, that is two separate PFCP establishments for the same subscriber on the same UPF.
Note: Co-located SGW-U + PGW-U when established by the same CPF (single PFCP Session Establishment) is supported.
- Third-party CPF or UPF.
- Limited support for PFCP IEs.
- PFCP PFD Management Procedure.
- PFCP Node Report Procedure.
- PFCP Overload Control Procedure.
- Co-located CPF and UPF in the same MCC cluster.
- Insertion strategy from SAEGW to SAEGW + SAEGW-U.
- GGSN service is not supported for CUPS.

Note: PGW with GGSN support enabled is supported for CUPS.

- Flow authorization interface is not supported for CUPS.

Unsupported TAM Features

- OSPF (Open Shortest Path First).
- TAM is not supported on the following products: GiGW, ePDG, TWAG, GTP Proxy.
- CUPS and non-CUPS UPF clusters do not support a mix of TAM and non-TAM sessions.
- Geo-redundancy.

Restrictions and Special Considerations

The following restrictions and special considerations apply to this release.

Restriction/Special Consideration	Description
The file /data/mysql/mysqlld.log does not rotate	When configured to rotate, the file /data/mysql/mysqlld.log does not rotate. In order for logrotate configuration to properly rotate the /data/mysql/mysqlld.log, you must issue the following command on each Acuitas server: <pre>mysql_config_editor set --login-path=client --user=root --host=localhost -password</pre> When prompted for a password, enter the MySQL root password. Repeat this procedure anytime you change the MySQL root password.
Do not use hypervisor-level commands to pause/unpause VMs	Do not use hypervisor-level commands to pause/unpause or suspend/resume virtual machines (VMs). Affirmed Networks does not support these commands because they may cause unintended events such as a cluster fault/reboot.
Rate limiting support	Rate limiting of up to 25% only of total proxy throughput is supported.
Acuitas support to manage an MCC with maximum pki ssh security level	Acuitas can manage a 9.2 or greater MCC with maximum pki ssh security level. However, Acuitas cannot manage an earlier version MCC with maximum pki ssh security level.
Data-usage statistics for CUPS sessions	For CUPS sessions, data-usage statistics, such as call-performance uplink/downlink packet/byte counts, deep-packet-inspection statistics, and any other data profile services statistics (for example, proxy, content-filtering, and other vas-service statistics) are available only on the UPF, not the CPF. On the CPF, these statistics are displayed as zero.
Signaling statistics for CUPS	All call performance statistics that pertain to GTP-C are available only on the CPF. On the UPF, these statistics are displayed as zero.

Restriction/Special Consideration	Description
An existing application-rule-group in MCC releases prior to 8.4 <i>cannot</i> be automatically converted to the MCC release 8.4 protocol-application-rule-group configuration	Due to architectural changes for the new DPI classification model in MCC release 8.4, an existing application-rule-group in MCC releases prior to 8.4 <i>cannot</i> be automatically converted to the MCC release 8.4 protocol-application-rule-group configuration. Important steps are required before and after upgrading from a release prior to 8.4 to the 8.4 or later release. - Before the upgrade, issue the following command to save the existing configuration: <pre>an3000-mcm-slot7cpu1# show running-config save test.cfg</pre> The configuration file is saved to the path <pre>./boot/partitions/current/home/users/admin/test.cfg</pre> - Perform the upgrade to MCC release 8.4. Be aware that MCC will experience loss of DPI functionality after the upgrade until you configure the new protocol-application-rule-group. - Following the upgrade, configure the new protocol-application-rule-group using the data in the test.cfg. Contact the TAC for assistance.
In configure_ha script, hostnames cannot contain upper case characters	Due to an anomaly with Red Hat Enterprise Linux 7, hostnames cannot contain upper case characters. Workaround: <i>For the configure_ha script, use only lower case characters in hostnames.</i> This lowercase hostname limitation does not apply for Standalone or Standalone Disaster Recovery node configurations.
Reporting an extremely large number of realtime statistics	When reporting an extremely large number of realtime statistics, Acuitas displays only the first 100 instances.
AI-vProbe Subscriber data plane mirroring limitation	A limitation exists in the support for AI-vProbe data plane mirroring in a cluster with compact building block strategy. When a vtap subscriber-tap is configured with reference-point set to core and HTTP Proxy is running on the CSM, packet replication occurs only for the egress packets. However, when HTTP Proxy is disabled, packet replication occurs as expected in both ingress and egress directions. There are no such restriction when HTTP Proxy is running on the ASM or the CPM.
MME network element name cannot contain a blank space	An MME network element name cannot contain a blank space.

Restriction/Special Consideration	Description
Implemented a change to the dpi-pre-detection setting	In MCC release 7.2.x, the workflow data-profile dpi-pre-detection-max-bytes and dpi-pre-detection-max-pkts default setting is changed to 0 (disabled): ■ dpi-pre-detection-max-bytes – Allow max number of bytes for DPI pre protocol and application detection ■ dpi-pre-detection-max-pkts – Allow max number of packets for DPI pre protocol and application detection If either of these objects were set to a non-zero value (enabled) in a prior release and you <i>do not</i> intend to use this functionality, you must manually set the value to 0 (disabled) prior to upgrading. Otherwise the system retains the non-zero value after the upgrade. For instructions, see Change to dpi-pre-detection Setting.
Clearing any of the DPI statistics clears all of the DPI statistics tables	Clearing any of the DPI statistics returns a confirmation prompt indicating that the action clears all of the DPI statistics. For example: <pre>zone <name> deep-packet-inspection statistics grid-level general clear</pre> This will clear all of the DPI statistics. Do you wish to proceed? (yes/no) : Any scripts that clear DPI statistics must account for this new prompt.
VoLTE SIP statistics are not supported on standalone SGW systems:	VoLTE SIP statistics are not supported on standalone SGW systems: <pre>an3000-mcm-slot7cpu1# show data-path statistics sip-statistics</pre>
VNF Manager scale in restriction	Do not use VNF Manager to scale in nodes other than the supported personalities of DCM/WSM/ASM.
vCenter requirement to add host resolution information in /etc/hosts	When hosts are identified by their names in vCenter, the VNF Manager must be able to resolve the host names to their IP addresses. The solution is to add IP address and hostname to the Acuitas /etc/hosts file.
vCenter datastore name must be unique	In vCenter, when you configure a datastore to create a VM, the datastore name must be unique. It cannot contain blank spaces or parentheses.
Geographic redundancy	Some geographic redundancy feature limitations exist in older releases when the MCC pair is at different major releases. When the paired MCCs run in this mode, Affirmed Networks recommends not making any configuration changes to the paired MCCs. The limitation no longer exists after all MCCs are upgraded to the same major release.
Georedundancy is not supported when upgrading from release 6.x to later releases, and 7.x/8.x to 9.x releases	Due to architectural and feature changes, georedundancy is not supported when upgrading from release 6.x to later releases, and from 7.x/8.x to 9.x releases. Both the primary and secondary MCC clusters must be upgraded to the current release to support georedundancy.
Online Software Upgrade (OSU)	OSU (for GiGw/TWAG/ePDG) from 7.2.x -> 8.1/8.2 is not supported due to architectural changes related to 7.3.x georedundancy features.

Restriction/Special Consideration	Description
User account lockout	If you attempt to log into Acuitas six consecutive times (or the number of times specified in the securityconfiguration.properties file), but fail because you supplied the wrong password, Acuitas automatically locks your account. You cannot log into Acuitas until your administrator unlocks your account. When the administrator unlocks your account, the failed attempts count is set to one. You have only one attempt to provide the correct password before Acuitas locks your account again.
The <code>emsha startall</code> script does not indicate that the script had completed successfully	When the script returns the success message: <code>Server started successfully</code> Press Enter to return to the operating system prompt.
Acuitas/MCC version compatibility	When Acuitas is running a version older than that of a managed NE, Acuitas treats the NE as if it is running the older version. All the NE's supported alarms, statistics, and configuration settings are those of the older version. For example, if the NE is running 5.3.8, but Acuitas is running 5.3.7, Acuitas treats the NE as a 5.3.7 system. If new statistical counters are added to an existing statistics table in the newer MCC release, that table may not be populated correctly when the Acuitas older version manages a newer NE. The same situation exists for new alarms; all new alarms are not handled properly. It is recommended that Acuitas be upgraded to the version matching the MCC at the earliest opportunity.
Warning that RSA host key changed due to a man-in-the-middle attack	If you receive a warning that the RSA host key changed due to a man-in-the-middle attack, after determining that no real man-in-the-middle attack occurred, you can recover from the error by following these steps: 1 Delete the fingerprints on <code>/root/.ssh/known_hosts</code>. 2 Generate a new set of keys by logging in as root and issuing the following command: <code>#ssh-keygen -t rsa</code> 3 Press Enter for each of the following prompts: Enter file in which to save the key (<code>/root/.ssh/id_rsa</code>): Enter passphrase (empty for no passphrase): Enter same passphrase again:
SGW Interim CDR triggers upgrade consideration	The SGW interface trigger template configures the events that trigger an interim (partial) record. Following an upgrade from a release prior to 5.0 to Release 5.0 or greater, you must reconfigure the offline charging interface (local and/or GTPP/Diameter) directly on the SGW. Reconfigure at Zone > <i>name</i> > Gateway > Sgw > <i>name</i> > Default Offline Interface

Restriction/Special Consideration	Description
Workflow data profile switching restrictions	The following workflow data profile switching restrictions apply to the QoS Policy. ■ Data profile switching does not operate properly in the case of network initiated /user initiated dedicated bearers (i.e. configured locally to create dedicated bearers from the MCC). ■ Data profile switching operates properly in the case of Dynamic bearers created by PCRF. ■ Data profile switching cannot convert a prepaid user to postpaid user. The system is unable to initiate a new diameter session due to switching workflows.
Benign Red Hat Messages	You can ignore the following Red Hat messages: <code>In order for Subscription Manager to provide your system with updates, your system must be registered with the Customer Portal. Please enter your Red Hat login to ensure your system is up-to-date.</code> <code>This system is missing one or more valid entitlement certificates. Please run subscription-manager for more information.</code>
Performance Manager – a real-time PM dashboard that includes both HTTP Proxy Transaction Success Rate and HTTP Proxy Transaction Success Rate % displays incorrect data	If you define a real-time PM dashboard that includes both HTTP Proxy Transaction Success Rate and HTTP Proxy Transaction Success Rate % together, the data plotted on the charts is not correct. Workaround: <i>View these KPI's in separate PM dashboards.</i>
HA – Warning during failover	As part of the database restore process or shutdown of the Acuitas server, the active server is shut down and the standby server becomes active. During the switchover, HTTP requests continue to be routed to the active server. As a result, Acuitas displays the following error in the client browser: <code>You are not authorized to perform this operation</code> This message is an interim warning when the server is not available during switchover. Once the standby server becomes active, you can refresh the browser to remove the warning.
Requirement to Change Password on Acuitas Server	If you change the password on the network element, you must also change that account information on the Acuitas server that manages that network element. To change the login password for the NE on the Acuitas server: 1 Right-click the network element and click Edit Element. 2 In the CLI & NETCONF section, enter the username and new password. 3 In the SFTP section, enter the username and new password. 4 Click Save. 5 At the confirmation prompt, click OK.

Restriction/Special Consideration	Description
BGP policies	BGP policies are used to control the routing information learned from a BGP peer and advertised to another BGP peer. If a BGP policy is modified, you must restart the peering session with the BGP peers to which this policy pertains. Under Ip Protocols > BGP[Vpn] > Router > <i>name</i> > Clear > All select the command (hard or soft) and click Select.
IP Route entity, its associated IP Interface entity, and its parent Network Context cannot be created within a single transaction	The NE has a limitation where the creation of an IP Route entity requires the preexistence of both its associated IP Interface entity and its parent Network Context. This limitation precludes creation of an IP Route entity, its associated IP Interface entity, and its parent Network Context within a single transaction. If an associated IP Interface entity and its parent Network Context entity are created within the same script, the user must commit the creation of the IP Interface entity and Network Context entity before the creation of the IP Route entity.
Do not use node number 1 for SSM deployments	Proxy traffic is dropped for some sessions by the Linux Kernel on the CSM if the SSM uses the network address 169.254.128.0 as its internal IP address. Workaround: <i>Redeploy the SSM and do not use node number 1 for SSM deployments.</i> To disallow setting SSM Node ID to 1: 1 Edit /opt/Affirmed/NMS/server/ems/resources/cloud.properties and set the following property: cloud.ssm.nonode1.enabled=true 2 Restart the Acuitas server.
Alarm and Event Logs View Archive feature is disabled for the release	Alarm and Event Logs View Archive feature is disabled for the release. Workaround: <i>Retrieve the archived alarms and event logs directly from the server in the /opt/Affirmed/NMS/server/ems/archive directory.</i>
Efficiently using the address space defined in your NAT pools	To efficiently use the address space defined in your NAT pools, Affirmed Networks recommends setting the max-nat-workflow-modules parameter to coincide with the number of SSMs/WSMs in your cluster. For example, in a cluster with only three SSMs/WSMs, set this parameter to 3 even though the parameter defaults to 15.

Restrictions for GQUIC User Agent Fingerprinting

- When an iOS device is connected to a primary device (Android) over Wi-Fi, it connects using proxy settings shared by the PdaNet application. As a result, all QUIC-generating applications on the iOS device revert to TCP. Therefore, GQUIC user agent fingerprinting is unable to detect tethered iOS devices.
- When an iOS device is connected to a primary device (Android) over Bluetooth, GQUIC traffic generated from the iOS device does not contain a UAID. Therefore, GQUIC user agent fingerprinting is unable to detect tethered iOS devices.

- Flow-level tethering detection is not supported for GQUIC user agent fingerprinting. Only the subscriber-level tethering status is detected by GQUIC.
- Tethering is not marked as true when the same UAID is present in GQUIC traffic generated from the same application on the primary and tethered devices. For example, if YouTube traffic from the primary device has GQUIC UAID as "Chrome/68.0.3440.91 Android 7.0; SM-G615F Build/NRD90M" and the same UAID is detected from the tethered device's YouTube traffic, GQUIC user agent fingerprinting cannot detect the tethering status.
- Tethering detection using GQUIC user agent fingerprinting requires GQUIC traffic to be generated by both the primary and secondary devices. For example, applications such as YouTube, Snapchat, Play Store, Google Drive, Google Search, and Google Maps should be used on both the primary and tethered devices to generate GQUIC traffic.

Configuration to Control DPI Decoding for Persistent HTTP Connections

For HTTP persistent connections, a single TCP connection is used to send multiple HTTP requests/responses. Because the persistent connection is not closed, all packets are subjected to DPI decoding, which can cause an increase in CPU usage.

To avoid a CPU usage spike, configure the following new parameter:

data-path dpi decode-first-http-transaction-only [enabled | disabled]

- **enabled** (Default) – On persistent HTTP connections, as soon as fastpath detects the first server response header, the connection is closed (at L7 level) and subsequent HTTP transactions *are not* sent to the DPI engine for decoding. This setting can cause classification errors, as DPI examines only the first transaction.
- **disabled** – On persistent HTTP connections, packets of all HTTP transactions are sent to the DPI engine for decoding.

Rename memimage to fault

Affirmed Networks renamed **memimages** to **fault**. This name change impacts filenames, traps, alarms, eventlogs, syslogs, and scripts used to capture faults and generate summary files and alarms. Below are examples of this renaming change:

Old Name	New Name
cluster <id> node <id> processes memimages	cluster <id> node <id> processes fault
Log Viewer => Memory Snapshot Files	Log Viewer => Fault Files
Save Tech Support Information => No Memory Snapshot Files	Save Tech Support Information => No Fault Files
infrastructure oom memimages	infrastructure oom faults
infrastructure memimages -filter generation	infrastructure faults -filter generation
infrastructure memimages generation	infrastructure faults generation
filecommand: category name for /public/ memimages	filecommand: category name for /public/ faults
file names under /var/ corefiles /* memimage *	file names under /var/ fault /* fault * <i>Note: Existing /var/corefiles/*core* files are not renamed, only new files use the new naming convention /var/corefiles/*fault*</i>
Alarm: anNew MemImage FileDetected	Alarm: anNew fault FileDetected
save-tech-support-info script change: -nomemimages	save-tech-support-info script change: --faults
show cluster <id> node <id> crash-detail < memimage -filename>	show cluster <id> node <id> memimage-detail < fault -filename>
show cluster <id> node <id> memimage -summary <seconds-since-epoch> < memimage -filename>	show cluster <id> node <id> fault -summary <seconds-since-epoch> < fault -filename>
show cluster <id> node <id> memimage -summary	show cluster <id> node <id> fault -summary
cluster <id> clear c-memimages	cluster <id> clear -faults
show cluster <id> node <id> memimage -detail	show cluster <id> node <id> fault -detail

Configuring VM Creation and Deletion Timeout

VNF Manager uses a default timeout timer for VM creation (30 minutes) and VM deletion (5 minutes). You can configure a different timeout, for example in a slow network (such as a network used to test the VNF Manager). It is not likely that a different timeout is required in a production network.

For VM creation, configure the stack creation timeout in minutes: Default 30, Minimum 1, Maximum 300.

For VM deletion, configure the stack deletion timeout in minutes: Default 5, Minimum 1, Maximum 300.

To configure these timeouts:

- 1 Use emsstop to shut down the Acuitas server:

```
cd /opt/Affirmed/NMS/bin
./emsstop
```

- 2 Edit /opt/Affirmed/NMS/bin/starttems.sh. After this line:

```
JAVA_OPTS="$JAVA_OPTS -Djboss.remoting.pooled-buffers=false"
```

add the following line:

```
JAVA_OPTS="$JAVA_OPTS -Dstack.create.timeoutMin=<timeout value>
-Dstack.delete.timeoutMin=<timeout value>"
```

- 3 Use starttems.sh to start the Acuitas server:

```
cd /opt/Affirmed/NMS/bin
./starttems.sh
```

Configuring Scale Timeout

VNF Manager uses a default timeout timer (120 seconds) after it disables a node before it removes the node during scaling in. You can configure a different timeout, for example in a slow network (such as a network used to test the VNF Manager). It is not likely that a different timeout is required in a production network.

To configure scale in timeout:

- 1 Change the value of the following property in /opt/Affirmed/NMS/server/ems/resources/cloud.properties. Make the change to both the active and standby server in an HA system.

```
cloud.mcc.disableWaitTimeSec=120
```

- 2 Use starttems.sh to start the Acuitas server:

```
cd /opt/Affirmed/NMS/bin
./starttems.sh
```

Change to dpi-pre-detection Setting

See the information in [Restrictions and Special Considerations](#) and then continue with this section.

Prior to upgrading from an MCC release prior to 7.2.x to Release 7.2.x or greater:

- If you are using the dpi-pre-detection functionality, set the value to a non-zero number prior to upgrading. The system retains that value after the upgrade. For example, set the value to 30 packets and 6000 bytes for a conservative configuration.
- If you *are not* using the dpi-pre-detection functionality, Affirmed Networks recommends setting the value to 0 (disabled).

After upgrading from an MCC release prior to 7.2.x to Release 7.2.x or greater:

- Settings that were previously unspecified or not present are automatically changed to 0 (disabled) by default.
- Settings that were previously specified with a non-zero value are retained.

The following table displays the behavior of these settings before and after upgrading to the current release.

Setting and Behavior Prior to Upgrade		Setting and Behavior After Upgrade	
Unspecified or not present (default)	Enabled by default and both packets and bytes are unlimited.	Values are set to 0 by default.	Functionality is disabled.
Both values are set to 0	Functionality is disabled.	Values prior to the upgrade are preserved and both values remain set to 0.	Functionality is disabled.
Non-zero value	Enabled with pre-detection max packets and bytes as specified in the configuration.	Values prior to the upgrade are preserved and both values remain set with non-zero values.	Enabled with pre-detection max packets and bytes as specified in the configuration.

Deploying the Acurtas Server in an IPv6 Network

To deploy the Acurtas server in an IPv6 network, perform the following tasks:

- 1 Use the qcow2 or ova image to configure the server IPv6 IP address.
- 2 Configure the Acurtas server with an IPv6 IP address:
 - For standalone server, run the `set_ipv6_address` script before starting the Acurtas server.
 - For standalone DR server, run the `set_ipv6_address` script before running the `configure_dr` script.
 - For HA, provide IPv6 IP addresses of the pair when running `configure_ha`.

Once these steps are performed, Acurtas server is run in IPv6 mode and should only be used to manage IPv6 NEs.

Historical GWGTPMESSAGESTATS Statistics

Historical GWGTPMESSAGESTATS statistics display the statistics type by its numerical value instead of its string value. Real Time GWGTPMESSAGESTATS statistics correctly display the statistics type's string value.

Historical Statistics Display

Real Time Statistics Display

GWGTPMESSAGESTATS

▼ numRx

☐ default_pgw_32

☐ default_pgw_33

☐ default_pgw_34

☐ default_pgw_35

☐ default_pgw_36

☐ default_pgw_37

[gwName=pgw,messageType=create-session-request]

[gwName=pgw,messageType=create-session-response]

[gwName=pgw,messageType=modify-bearer-request]

[gwName=pgw,messageType=modify-bearer-response]

[gwName=pgw,messageType=delete-session-request]

[gwName=pgw,messageType=delete-session-response]

Use the following mappings to determine the string value for each statistics type:

Numerical Value	String Value
32	create-session-request
33	create-session-response
34	modify-bearer-request
35	modify-bearer-response
36	delete-session-request
37	delete-session-response
38	change-notification-request
39	change-notification-response
64	modify-bearer-command
65	modify-bearer-failure-indication
66	delete-bearer-command
67	delete-bearer-failure-indication
68	bearer-resource-command
69	bearer-resource-failure-indication
70	downlink-data-notification-failure-indication

Numerical Value	String Value
71	trace-session-activation
72	trace-session-deactivation
73	stop-paging-indication
95	create-bearer-request
96	create-bearer-response
97	update-bearer-request
98	update-bearer-response
99	delete-bearer-request
100	delete-bearer-response
101	delete-pdn-connection-set-request
102	delete-pdn-connection-set-response
166	create-indirect-data-forwarding-tunnel-request
167	create-indirect-data-forwarding-tunnel-response
168	delete-indirect-data-forwarding-tunnel-request
169	delete-indirect-data-forwarding-tunnel-response
170	release-access-bearers-request
171	release-access-bearers-response
176	downlink-data-notification
177	downlink-data-notification-ack
200	update-pdn-connection-set-request
201	update-pdn-connection-set-response
1003	version-not-supported-indication
1016	create-pdp-ctxt-request
1017	create-pdp-ctxt-response
1018	update-pdp-ctxt-request
1019	update-pdp-ctxt-response
1020	delete-pdp-ctxt-request
1021	delete-pdp-ctxt-response
1022	initiate-pdp-ctxt-activation-request

Numerical Value	String Value
1023	initiate-pdp-ctxt-activation-response
1027	pdu-notification-request
1028	pdu-notification-response
1029	pdu-notification-reject-request
1030	pdu-notification-reject-response

Scale In Limitations

Scale In has the following limitations:

- On Scale In of a DCM, some sessions that do not passively get closed (on the gateway that is being removed) will be closed, disrupting the traffic belonging to those sessions.
- On Scale In of an ASM, some HTTP connections that do not close during the dry-up period will be closed. The UE can re-establish a new connection.
- When a WSM node is disabled as part of dynamic scaling, the grid removes the range of FSAPIs that were earlier assigned to the card. The gateway migrates all of the sessions using FSAPIs of this card to another active SSM (or WSM) node. As a result, workflow sessions are deleted on an old FSAPI and a new workflow session are created with the new FSAPI. The following limitations arise with this implementation:
 - During the session migration from an old FSAPI to a new FSAPI, data may be lost in the timing window between when the session is deleted on the old FSAPI and recreated on the new FSAPI.
 - If Proxy is enabled, all of the connection data information maintained by wfdivert module will be lost and the corresponding HTTP transactions will timeout. The same limitation applies to all of the Content services that use this module (SIP, WAP Gateway, SMTP Proxy).
 - If NAT is enabled, all of the active connections (not just TCP connections) will be lost. They must be reestablished after the workflow session is recreated on the new WSM. The same limitation applies to all the supported ALGs, as the flow state is not synchronized with a standby (SIP, RTSP, FTP, TFTP ALGs).
 - When an NE is deployed as a PGW or GGSN, the reassembly of fragmented packets that originate from UE is performed in the workflow manager module. As part of dynamic contraction, when a session is migrated to another SSM (WSM), any fragments that have not yet completed reassembly will be lost (because after FSAPI migration, the workflow for this session is anchored on a different WSM.)
 - The grid can choose to use WSM modules as backup for FSAPIs assigned to other cards. When a WSM is Scaled Out, the grid designates one of the remaining cards to work as a backup for those FSAPIs for which the WSM being disabled was serving as backup. Thus, the backup for certain FSAPIs will move from one node to another during both dynamic WSM expansion and contraction. This condition exposes a timing window if (even before the workflow sessions on the backup FSAPIs are fully set up) the SSM(WSM) with the active FSAPI either reboot or crash. Data will be lost during this timing window. This condition is expected to be transient, as normal operation is expected to resume as soon as the workflow session is created on the SSM(WSM).

Acuitas Software Installation/Upgrade

Affirmed Networks supports Acuitas server installation/upgrade on an Affirmed Networks supported system.

For instructions on installing Acuitas server software, see the *Acuitas System Administration Guide*.

Support for Acuitas Software Version with Red Hat/MySQL Image

The following tables list the Acuitas releases that can be used with the various Red Hat/MySQL images (OVA and qcow2).

	an-v1.0-rhel-server-6.8-x86_64-mysql-5.6.30	an-ems-v1.2-rhel-server-6.8-x86_64-mysql-5.6.31	an-ems-v1.1-rhel-server-6.8-x86_64-mysql-5.6.34	an-ems-v2.1-rhel-server-7.5-x86_64-mysql-5.7.23.ova an-ems-v2.0-rhel-server-7.5-x86_64-mysql-5.7.23.qcow2
Acuitas 6.5.x or earlier	Yes	Yes	No	No
Acuitas 6.6.x	No	Yes	No	No
Acuitas 7.x	No	Yes	Yes	No
Acuitas 8.x	No	No	No	Yes
Acuitas 9.0.x - Acuitas 9.3.x	No	No	No	Yes

Important: Acuitas 9.4 can be used only with these Red Hat/MySQL images:

- an-ems-3.0.5-rhel-7.7-mysql-8.0.18.iso
- an-ems-3.0.6-rhel-7.7-mysql-8.0.18.ova
- an-ems-3.0.5-rhel-7.7-mysql-8.0.18.qcow2

Requirement to Upgrade the OS

If you are upgrading from a release prior to 9.4.0.0 to release 9.4.0.0 or later, that upgrade requires an upgrade to the OS. For instructions on how to perform the OS upgrade, see *OS Upgrade Considerations* in the *Acuitas System Administration Guide*.

Requirement to Install Acuitas Packages for VMware

If the VNF Manager is used for MCC lifecycle management with vCenter, an extra step is needed during Acuitas upgrade. Following the Acuitas upgrade, you must perform the following procedure. Even if you had previously installed the packages, you need to reinstall them.

To install Acuitas packages for VMware:

- 1 Log into the Acuitas server as `root`.
- 2 Issue the following commands to untar the packages tarball:

```
cd /opt/Affirmed/NMS/ctool  
tar -zxvf ./acuitas-commission-tool.tar.gz
```
- 3 Issue the following commands to install the packages:

```
cd acuitas-commission-tool  
./install.sh
```
- 4 Accept the VMware license agreement.
- 5 When prompted to install the RedHat pre-built packages, enter **yes**.
- 6 Issue the following command to remove the acuitas-commission-tool directory created by the tar command:

```
cd ../; rm -rf acuitas-commission-tool
```

Known Anomalies

The following known anomalies exist in this release.

New Major Anomalies

The following major anomalies exist in this release.

Major Anomaly	Description
45741 CUPS Manager — PFCP Profile drop down menu does not function	When creating a CUPS Group from the CUPS Manager, if you select an existing PFCP Profile from the drop down menu, the values are not populated. Workaround: <i>When creating a CUPS Group from the CUPS Manager, manually enter the PFCP Profile information.</i>
46939 VNF Manager cannot scale out a renamed cluster	VNF Manager cannot scale out a renamed cluster. Workaround: Perform these steps from the VNF Manager: 1 Export and save the cluster configuration. 2 Remove the cluster from the VNF Manager. 3 Re-import the cluster configuration, but give the cluster the new name. 4 Wait for a few minutes, refresh the cluster, and verify its state is synchronized. 5 Perform with desired scale out.
47221 Subscriber Analyzer shows only one key instead of all provisioned keys	Acuitas does not list all the configured Subscriber Analyzer keys. Workaround: <i>Use the CLI to view the complete Subscriber Analyzer information:</i> show running-config workflow subscriber-analyzer <name>

Major Anomaly	Description
47396 Network View does not display any NEs	After changing the IP Address of Acuitas server nodes or after restoring the database (with different Acuitas node IP address) in a HA setup, the following issues occur: ■ Network View does not display any NEs ■ Acuitas nodes do not get updated with new IP address Workaround: Perform these steps: 1 Stop the standby Acuitas server. 2 Run the following queries on the active Acuitas node to delete Acuitas node entries in RESOURCECOMPONENT, EMSNODE, and MANAGEDCOMPONENT: <pre>mysql -uroot -p<root password> AFFIRMED_NMS_DB; mysql> delete from RESOURCECOMPONENT where parentcomponentid=0 and resourceid=1; mysql> delete from EMSNODE; mysql> delete from MANAGEDCOMPONENT;</pre> 3 Restart active Acuitas server and verify that the issue is resolved. 4 Start the standby Acuitas server. Note: As this workaround resets the Acuitas node entries, any collected Acuitas node statistics (EMS_STATS, DISK_STATS) are lost.

Existing Critical Anomalies

The following critical anomalies found in an earlier release also exist in this release.

Critical Anomaly	Description
45709 Geographic Redundancy Group Configuration Compare and Replicate does not function with 9.3 MCCs	The tool to compare and replicate configuration data on the active and standby nodes of a Geographic Redundancy group does not function with 9.3 MCCs.
43623 DPI statistics and QCI/ARP statistics do not support thresholds	Acuitas does not support threshold association for Pace 1 DPI statistics and QCI/ARP statistics.
36317 Database Restore takes a long time to restore	Database Restore takes a long time (30-60 minutes) to restore. Workaround: Do not cancel the restore process; let it continue until completion.

Existing Major Anomalies

The following major anomalies found in an earlier release also exist in this release.

Major Anomaly	Description
46970 Acuitas DB backup fails when the MySQL password contains special characters	Acuitas DB backup fails when the MySQL password contains special characters
45630 Standalone DR - replication alarm is generated but is not shown in Acuitas	When using a non-default interface (not eth0) on a standalone DR system, the replication alarm is generated but is not shown in Acuitas. Workaround: As root, change the third line in the affirmed cron file to the following: <pre>*/5 * * * * affirmed /opt/Affirmed/NMS/bin/ems_sa_monitor --ipinterface <interface name> > /dev/null 2>&1</pre>
45427 HA-DR - No alarm is raised if either of slave pair is down	HA-DR - No alarm is raised if either of a slave pair is down. The alarm emsDBReplicationDown should be raised.
42171 The DPIGENERAL KPI CSV file is not generated	The DPIGENERAL KPI CSV file is not correctly generated when any one of the following files is not present: ■ EMSDPIPROTOCOLSGRID ■ EMSDPIAPPLICATIONSGRID ■ EMSDPIOPERATINGSYSTEMSGRID
42987 VNF Manager does not correctly calculate the KPI ASM Throughput Capacity Utilization. VNF Manager uses the KPI to define dynamic scaling policy	The ASM Connection Utilization is not calculated correctly. As a result the dynamic scaling based on ASM Connection Utilization cannot be used.
43863 VNF Manager does not display correct status after creating or failing to create a stack	When VNF Manager creates/fails to create a stack for either MCC and MCC MME, it does not display the correct status. Instead, the status is blank in the status bar.
43464 Geographic Redundancy Manager - cannot source configuration on secondary C-Plane	Attempts to source the configuration from the primary C-Plane to the secondary C-Plane fail because the <PGW/SGW> -> PFCP endpoint must be committed on the device before the <PGW/SGW> -> PFCP peer can be configured. Workaround: Before replicating the configuration from the primary C-Plane to secondary C-Plane, manually configure the PGW/SGWs on the secondary C-Plane that has the configured PFCP peer and PFCP endpoint.

Major Anomaly	Description
42231 SCEF alarms list the SCEF trap OID in the Category field, instead of the SCEF trap name	SCEF alarms list the SCEF trap OID in the Category field, instead of the SCEF trap name. Workaround: Consult the SCEF MIB to map the trap OID to the trap name.
40731 Configuration Navigator — Unable create GTP interfaces on an SSF device	The SSF device requires you to configure/delete GTP Access interface and GTP Network interface simultaneous as a pair, which is not supported by the Configuration Navigator. Workaround: Use the CLI to configure the GTP interfaces.
39760 In certain situations, VNF Manager cannot heal a VM	In certain situations, a host hosting a VM becomes non-functional and cannot be managed by the OpenStack controller. Although the user can heal the VM, the previous host is not re-commissioned and managed by the OpenStack controller without re-initialization (such as cleaning up of all its previous VM configuration.)
37878 Attempting to upload a URL List that exceeds the maximum size fails without an error	Attempting to upload a large file for URL List that exceeds the maximum size fails with an error. Workaround: To determine if the error occurred, view the /opt/Affirmed/NMS/server/ems/log/affirmed.log file.
33953 Using emsupgrade script to upgrade Acuitas server causes a benign error	Using emsupgrade script to upgrade Acuitas server causes a benign error when stopping the Acuitas server. The Acuitas server is correctly stopped. You can ignore the benign message.
33404 Geographic Redundancy Manager - Last Successful Switchover column does not display the time of the last successful switchover	In the Geographic Redundancy Manager, the Last Successful Switchover column does not display the time of the last successful switchover. Workaround: Click the View Chronological Switchover Log icon to view the switchover events of each Geographic Redundancy group.
30652 User should be restricted from performing some actions on an unmanaged NE	When an NE is Unmanaged, the user should be restricted from performing the following actions: ■ Running a software schedule on the NE ■ Using NE Configuration Backup and Restore to restore NE's configuration data
30110 Following an upgrade, existing user-defined Dashboard filters do not persist	Following an upgrade, existing user-defined Dashboard filters do not persist.
27116 Scaling out a node using the VNF Manager and Provisioning REST API does not log the operation in the Acuitas configuration audit log	Scaling out a node using the VNF Manager and Provisioning REST API does not log the operation in the Acuitas configuration audit log.

Major Anomaly	Description
23846 Search/Paging feature does not function correctly in Configuration Navigator	The Search/Paging feature does not function correctly in the Configuration Navigator. When you enter a string in the Search text box and click the Search icon, Acuitas lists the correct search results. However, paging through the search results does not function correctly.
23802 Workflow - Dragging and dropping objects from Available List to Selected List does not remove the instance from the Available List	Moving an instance from the Available List to the Selected does not remove the instance from the Available List. This issue is a display issue only, no other problems occur as a result of the incorrect display.
22431 Error when synchronizing a workflow to another device	Attempts to synchronize a workflow when both the source and destination nodes have the same workflow with different content fails to overwrite the destination workflow with the source workflow and returns the following error: Exception while trying to create objects
22773 Affirmed Open Workflow application does not support specifying a Geo Redundancy Group to bind to a Subscriber Analyzer	Using the Affirmed Open Workflow application, the user cannot specify a Geo Redundancy Group to bind to a Subscriber Analyzer. Workaround: Use the Configuration Navigator to specify a Geo Redundancy Group to bind to the Subscriber Analyzer.

Existing Minor Anomalies

The following minor anomalies found in an earlier release also exist in this release.

Minor Anomaly	Description
35552 Following an HA+DR upgrade, re-configuring the pair as DR slaves fails when using the following command: ./configure_dr convert slave	Following an HA+DR upgrade, re-configuring the pair as DR slaves fails when using the following command: ./configure_dr convert slave Workaround: Following the upgrade, use the following command instead: ./configure_dr convert slave --masterhost <masterhost>

Problems Fixed

This section lists the corrections in this release.

Corrections in Release 9.4.0.0

Anomaly	Description
40464 Alarm Forwarding of vCenter alarms are missing OID values when forwarded manually to northbound interface	Alarm Forwarding of vCenter alarms (vCenterOldStatus and vCenterNewStatusa) re no longer missing OID values when forwarded manually to northbound interface.
44655 Real Time statistics report does not function correctly for SCEF tables	Real Time statistics report now functions correctly for the following SCEF tables: ■ SCEFAPIGW AGGREGATEPERFSTATSCHASSIS ■ SCEFDNSRESOLUTIONAGGREGATESTATS
46643 Importing a PM Dashboard CSV file into the PM Dashboard does not import the first few PM Dashboards correctly	Importing a PM Dashboard CSV file into the PM Dashboard imports all the PM Dashboards correctly. None of the Dashboards are displayed as blank in the PM Dashboard tab.
46884 Acuitas fails to replicate CRBN Profile from the C-Plane to the U-Plane	Added CRBN Profile to the list of common configuration to be replicated from the C-Plane to the U-Plane.
46972 User with administrator rights cannot change the password for other users without providing the current user password	Administrator users can now change other users' passwords without having to provide the current password.
47141 Missing NEs in user-triggered NE Backup and Restore	Acuitas now handles invalid version values and lists the NE's correctly.
47292 configure_dr can sometime fail due to MySQL authentication failure using environment variables	Acuitas scripts now requires the user to use mysql_config_editor to store MySQL login path to be used for MySQL authentication.
47365 Alarms are not archived in /opt/Affirmed/NMS/server/ems/archive/alarms_export	Alarm archives are correctly written to the alarm archive directory, not the event log directory.

Anomaly	Description
47507 DPIWEBHTTP TRANSACTIONSGRID table is missing from Performance Manager statistics collection	Performance Manager now includes the DPIWEBHTTPTRANSACTIONSGRID table: ■ serverGetResponses ■ serverGetServiceTimeAvg ■ serverGetServiceTimeMovingAvg ■ proxyGetResponses ■ proxyGetServiceTimeAvg ■ proxyGetServiceTimeMovingAvg
47546 PROXYBUFFERSTATS SLOT table is missing vm-hostname	Performance Manager includes the vm-hostname in exported 5-minute PM CSV files for the PROXYBUFFERSTATSSLOT statistics table.
47570 CONTENTHTTPPROXY TCPOPTZSTATSSLOT table is missing vm-hostname	Performance Manager includes the vm-hostname in exported 5-minute PM CSV files for the CONTENTHTTPPROXYTCPOPTZSTATSSLOT statistics table.

Corrections in Release 9.3.3.0

Anomaly	Description
47007 Database backup in Acuitas 9.3.2.0 HA does not function correctly	Database backup in Acuitas 9.3.2.0 HA now functions correctly.

Corrections in Release 9.3.2.0

Anomaly	Description
46465 System KPI values for ISMs/SSMs showed high CPU utilization. The System KPI dashboard used SLOT_STATS statistics, which can be misleading	Added the statistics from the following tables to the Dashboard Metric Configuration screen: SYSTEMCPUSTATISTICS ■ Average 1 min CSM CPU Memory Utilization ■ Average 1 min MCM CPU Memory Utilization ■ Average 1 min SSM CPU Memory Utilization ■ Average 5 min CSM CPU Memory Utilization ■ Average 5 min MCM CPU Memory Utilization ■ Average 5 min SSM CPU Memory Utilization ■ Average 15 min MCM CPU Memory Utilization ■ Average 15 min CSM CPU Memory Utilization ■ Average 15 min SSM CPU Memory Utilization SYSTEMMEMORYSTATISTICS ■ Average 1 min CSM CPU Memory Utilization ■ Average 5 min CSM CPU Memory Utilization ■ Average 15 min CSM CPU Memory Utilization ■ Average 1 min SSM CPU Memory Utilization ■ Average 5 min SSM CPU Memory Utilization ■ Average 15 min SSM CPU Memory Utilization ■ Average 1 min MCM CPU Memory Utilization ■ Average 5 min MCM CPU Memory Utilization ■ Average 15 min MCM CPU Memory Utilization You can create a Dashboard that uses these statistics, instead of the SLOT_STATS.
46368 In an HA deployment, after changing the replication password, the database backup failed from Acuitas	Database backup now uses the configured mysql password to access the database.
45817 Performance Manager does not return values for some statistics for the Real Time Report	The following statistics are now retrievable by the Real Time Report. SLOT_STATS Isolated CPU Utilization Normalized General CPU Utilization
44502 When TACACS was unreachable, a timeout caused MCC to become unmanaged by Acuitas	To correct the situation, the SSH timeout for Netconf has been increased to 60 seconds.

Corrections in Release 9.3.1.0

Anomaly	Description
46212 After an Acuitas HA+DR recovery, mysql database on the standby was out of sync	This fix relaxes the sync settings to improve MySQL replication performance.
45897 Purge process can crash when attempting to purge a large number of CLEARED alarms	Acuitas uses purge settings to periodically purge the following alarms: ■ Cleared alarms, regardless of whether the alarms have associated event logs ■ Any event log mappings associated with the alarms In addition, added Acutas support to Purge All cleared alarms and Purge All event logs: Fault > Alarms > Purge All Fault > Logs > Purge All
45850 Configuration Navigator - Cluster > Core Layout does not display data	Cluster > Core Layout now correctly displays data.
45687 Some users logins were locked out due to DORMANCY_PERIOD_EXCEEDED, even though the user had logged in recently and not exceeded the dormancy period	This fix preserves the last login time during the user detail update process.
45663 CUPS Manager does not remove the configuration from the MCC following a CUPS Group creation failure	When the user attempts to create a CUPS group from the CUPS Manager, if an exception occurs while committing the configuration on the User Plane, the configuration on the Control Plane is not removed. Acuitas raises the critical alarm: CUPS Group creation failed from the EMS. C-Plane configuration on <cluster> has to be manually deleted You must delete the CUPS configuration manually from the Control Plane and manually clear the alarm.

Anomaly	Description
45421 During specific points in several scripts, closing the terminal can impact the script process	Changed the behavior of the following scripts that were previously run in the background yet still output to the screen, where the user could enter <CTRL>-C to break out of the script, but still leave it running in the background. ■ configure_ha ■ configure_dr ■ emsupgrade ■ haupgrade Now, after initiating the script and accepting/confirming all questions, the process is started completely in the background and the user is prompted to tail the relevant log file for progress: The command \$COMMAND is running in the background. Check log file \$LOG_FILE for the command result.
42852 Configuration Navigator does not alert user that the cluster immediately reboots when TAM Policy Mode is changed from disabled to enabled-for-cluster	Changing TAM > Policy > Mode from disabled to enabled-for-cluster now alerts the user that the change triggers an immediate cluster reboot: ALERT: About to trigger an immediate cluster reboot! Click OK. ■ To initiate the cluster reboot, click the Save icon.  ■ If you do not want the cluster to reboot, change the setting back to disabled and click the Save icon. 
45844 CUPS Manager - Auto Refresh does not function on CUPS Manager	You no longer have to refresh the page to populate any operation performed on CUPS Manager.
45822 CUPS Manager - changing audit interval can cause auto audit to fail	When you use CUPS Manager to change the audit interval, audit now correctly populates the result for the CUPS group.
45694 CUPS Manager - CUPS group does not clean up once replication is performed for both C-plane and U-plane	CUPS Group mismatch icon is correctly removed and the group color changes to grey once replication is performed on both C-plane and U-plane (with no mismatch).
45300 CUPS Manager should raise an alarm when a CUPS group mismatch status changes	When CUPS auto auditing is enabled, with each discovery/audit, CUPS Manager maintains a mismatch count for each CUPS group. Acuitas now generates the following critical alarm when a CUPS group status changes from matched to mismatched and clears the critical alarm when a CUPS group status changes from mismatched to matched: CUPS configuration mismatch found between <cluster 1> and <cluster 2> in CUPS group <CUPS group name> Remedy: Perform sync in CUPS Manager

Corrections in Release 9.3.0.0

Anomaly	Description
45820 Acuitas HA cluster cannot survive consecutive switchovers	More than one switchover of an HA cluster no longer causes the new master to not own the VIP and the heartbeat process to not run.
45814 Following a password change for emsadmin user on MCC, VNF Manager on OpenStack fails to scale out a node	VNF Manager now uses passwords from registered MCCs instead of default passwords for scaling out in OpenStack environment.
45449 Unable to delete MCC network element	Deleting an MCC network element in a specific state no longer returns an error.
44670 When the system automatically unmanages a device due to authentication failure, the operation is not logged in audit logs	This fix adds support to: ■ Log system triggered (automatic) device unmanaged state. ■ Generate the following critical alarm for system triggered (automatic) Un-managed action and clear those alarms when the device is moved to managed state: Device has been automatically unmanaged due to NETCONF authentication failure

Corrections in Release 9.2.1.0

The following anomalies are corrected in Release 9.2.1.0.

Anomaly	Description
42000 Partial data for multi-table KPIs from any MCC may cause a memory leak that can accumulate and eventually cause the process to stop due to out of memory error	To avoid a memory leak for multi-table KPIs, at any point, Acuitas stores only the latest 6 intervals (30-minute).
43929 Acuitas HA deployment may miss the option to configure LDAP	If a deployment that was upgraded to EMS 9.2 is missing the LDAP configuration option, use the Workaround below. For other deployments, the LDAP configuration option now appears post upgrade to EMS 9.2.1. Workaround: Follow these steps: 1 Extract the AuthenticationServer.xml from the emsha rpm. For example, if the emsha rpm is located in /data/affirmed: <pre>cd /data/affirmed mkdir temp cd /temp rpm2cpio ../emsha-xxx.rpm cpio -idmv "*AuthenticationServer.xml"</pre> 2 Once you see the command output list the file /opt/Affirmed/NMS/server/ems/conf/AuthenticationServer.xml, exit out the command. The file should be located in that path under the temp directory. 3 Copy the file to /opt/Affirmed/NMS/server/ems/conf/AuthenticationServer.xml on both the active and standby Acuitas servers.
43994 Acuitas should allow user to change MySQL replication password	Acuitas now supports allowing the user to change MySQL replication password.
43995 User cannot edit an existing template maintenance file	Editing an existing template maintenance file no longer return an error. With this fix, Acuitas properly handles the save template response validation on the client side and closes Template Maintenance window after updating the template details.
44015 User-defined templates created before the upgrade cannot be edited/deleted/obsoleted after the upgrade	User defined Templates created before the upgrade can now be edited/deleted/obsoleted after the upgrade.

Anomaly	Description
44020 Authentication using IPv6 LDAP IP address does not function correctly	Configuring IPv4 or IPv6 LDAP IP addresses through configureserver.sh script now functions correctly.
44399 Authentication using IPv6 LDAP IP address does not function correctly when SSL is configured	LDAP authentication now succeeds for IPv6 IP address when SSL is enabled

Corrections in Release 9.2.0.0

The following anomalies are corrected in Release 9.2.0.0.

Anomaly	Description
41687 DPI KPIs in the PM Dashboard do not create an export report, even when data shows in the PM Dashboard	Daily export of EMS DPI KPI Dashboard now succeeds when the PM dashboard includes both the % (Application/Protocol/OS) Network Usage and the (Application/Protocol/OS) Network Usage (Mbps) together.
43361 PM statistics collection may fail due to an unexpected file in staging directory	While creating staging directories, the system removes possible collision files to allow statistics collection to proceed without error.
43339 VNF Manager and Provisioning REST API executeEmsTemplate command fails when executed against a SCEF device	The executeEmsTemplate command now correctly executes for SCEF.
43329 Acuitas failed to add an MCC Network Element	The device addition process now uses the correct default software entry and validation.

Anomaly	Description
42928 Acuitas PM Dashboard reports exceed 100% success on UDP-Proxy transactions	This issue existed because the attempted statistic was updated at the beginning of the flow and the final statistic was updated at the end. Because Acuitas captures the data in 5 minute intervals, some statistics updates were allowed to span two or more intervals, skewing the calculations. Now, the update for the statistic occurs when the flow is closed, so that the initial and final statistics updates fall in the same 5 minute interval. In addition, the UDP-Proxy - Transaction Success Percentage KPI formula has changed to: $\text{UDPPROXYSTATSSUMMARYCHASSIS.num_udp_proxy_summary_successes} / (\text{UDPPROXYSTATSSUMMARYCHASSIS.num_udp_proxy_summary_successes} + \text{UDPPROXYSTATSSUMMARYCHASSIS.num_udp_proxy_summary_aborts} + \text{UDPPROXYSTATSSUMMARYCHASSIS.num_udp_proxy_summary_failures}) * 100$
42410 PM Dashboard with the same metric in two slots each with a different configuration only exports the first report	Now, when a PM Dashboard has the same metric in two slots each with a different configuration (such a Show Top, Bottom, All and Group By, Aggregate, Filter By), both reports export correctly.
42469 Instantiating SCEF cluster from VNF Manager caused an error	Removed the tab ('\t') character from the heat-template file to allow instantiation.
43824 Attempts to configure and save a maintenance template file (Configuration > Template > Maintenance) caused a Failure Error message	The maintenance template addition process now uses the correct default software entry and validation.
43220 Alarm Triggered Configuration did not function correctly	Added restrictions to the Template Maintenance window to prevent the user from modifying/deleting/obsoleting the default system-defined templates. Added restrictions to the Alarm Triggered Configuration window to prevent the user from modifying/deleting the default system-defined configuration publisher details or its filter associations. Added View Configuration Publisher Details window. During DB upgrade, wrong mappings in the FILTERCONFIGPUBLISHERMAPPING table were corrected.

Corrections in Release 9.1.2.1

The following anomalies are corrected in Release 9.1.2.1.

Anomaly	Description
42368 Configuration Navigator does not display parameters for a saved Service Construct > Code Mapping List instance	Configuration Navigator now correctly displays parameters for a saved Service Construct > Code Mapping List instance.
42507 LTEMME_TAU_KPI KPI CSV File and PM dashboard do not correctly calculate the KPIs.	LTEMME_TAU_KPI KPI CSV File and PM dashboard now correctly calculate the KPIs.
42484 LTEMME_ITAU_KPI KPI CSV file and PM dashboard do not correctly calculate the KPIs.	LTEMME_ITAU_KPI KPI PM dashboard now correctly calculates the KPIs.
42559 Incorrect status listed in the CSV Stats Recovery > View Details window	When an MCC is not reachable, the View Details window now correctly lists status as Not Reachable.

Corrections in Release 9.1.2.0

The following anomalies are corrected in Release 9.1.2.0.

Anomaly	Description
40474 Configuration Navigator — Unable create and save an Imsi Range List	You can now create and save the following object: Service Construct > Imsi Range List
42367 vCenter validation messages list incorrect result for SR-IOV host.	This issue was introduced in Acuitas 9.1.1.0. The validation messages are now correct and display the same content as for the pre-Acuitas 9.1.1.0 release.
42520 Scaling out a VM using VNF Manager causes an error	VMware ovftool package are upgraded from 4.1.0 to 4.3.0 in acuitas-commission-tool.tar.gz. This package is installed when installing ovatoools. Important: If the VNF Manager is used for MCC lifecycle management with vCenter, you must install the Acuitas packages for VMware following the Acuitas upgrade. For instructions, see Requirement to Install Acuitas Packages for VMware .

Anomaly	Description
42531 Due to errors in processing the source field of MCC traps, sometimes the same MCC alarm is re-raised and remains an alarm even after MCC clears it	The source field is set as the value of source varbind, as defined in PARSERS table, instead of MCC IP address, and DG traps are processed separately.

Corrections in Release 9.1.1.0

The following anomalies are corrected in Release 9.1.1.0.

Anomaly	Description
38507 IPv6 HA sometimes fails to recover from a split-brain scenario	IPv6 HA no longer fails to recover from a split-brain scenario.
40683 Acuitas Geographic Redundancy Manager replication failed because it replicated standalone objects referred by other Geo-redundancy entities	Acuitas Geographic Redundancy Manager no longer replicates the standalone objects.
40709 Some Acuitas screens do not render properly	Acuitas now pre-compiles missing JSPs and adds proper entries in web.xml.
40810, 40821 Configuration Navigator does not populate statistics on an SSF device	Configuration Navigator now populates statistics for the following SSF tables: ■ SSF_GTP_3GPP_V1_LEFT ■ SSF_GTP_3GPP_V1_RIGHT ■ SSF_GTP_3GPP_V2_LEFT ■ SSF_GTP_3GPP_V2_RIGHT
40833 CUPS — Directories on the Control Plane are not replicated to the User Plane	The following configuration directories on the Control Plane are now replicated from the Control Plane to the User Plane. ■ service-construct content-insertion - /tosdb/content/insertion ■ service-construct ca-certificate-list - /tosdb/content/tls
40845 Unable to execute Scale-In/Out of MCC VMs after the emsadmin password is changed	When communicating with MCC during Scale-In/Out, VNF Manager now uses the login credentials from the registered MCC.
40864 Following a switchover from Active to Standby, SNMP Community String does not synchronize on a secondary server	Following a switchover from Active to Standby, the SNMP Community String now synchronizes on the secondary server that became active.

Anomaly	Description
41110 CUPS — Configuration object on the Control Plane is not replicated to the User Plane	The following configuration object on the Control Plane is now replicated on the User Plane during the replication process. ■ services http-proxy ltmi
41117 Heat templates do not configure security-groups for non-base ports	Heat templates configures security-groups for Management ports of CSM and SSM nodes.
41426 Acuitas Historical Statistics do not display instance identifiers for the LTEMME_PagingProfile statistics table	Historical Statistics now display instance identifiers for the LTEMME_PagingProfile statistics table.
41433 Acuitas cannot manage MCC MME device when device full version is too long	Added support to extract the proper version of MME device.
41450 CSGN VM instantiation fails when the user tries to instantiate MME with the same logical IP address as one already instantiated	VNF Manager now checks if the logical IP address is already used before instantiation to avoid this user error.
41487 VNF Manager - Configuring more than two ISM or WSM VMs in a cluster fails during instantiation	VNF Manager now configures a max-asm value during mcc instantiation that specifies the maximum number of ASMs, ISMs, and WSMs in the cluster.
41552 VNF Manager does not support deploying a VM with SCSI disk type	To automate VNF Manager SCSI support, the administrator must install ovatools-1.3.4. To install an-mod-ova version 1.3.4, as root , issue the following commands: <pre>cd acuitas-commission-tool ./install ovatools-1.3.4</pre>
41679 The incorrect NumCreateSessionReject ApnCongesteCritical statistic (misspelling) is a duplicate of NumCreateSessionReject ApnCongestedCritical and appears in exported csv files	The incorrect statistics is deprecated in MCC 9.2 and greater but is still visible in early versions.

Anomaly	Description
41729 Configuration Navigator for MME > Routing Instance > Instance object does not allow you to configure IP addressing for LB/DATA/EDP VMs, causing an error when attempting to save the configuration	Configuration Navigator now allows you to configure Lb Instance, Data Instance, and Edp Instance required objects.
42093 CUPS — Replication causes User Plane to crash and keep booting	The replication from Control Plane to User Plane no longer causes User Plane to crash because the auto index values are now not replicated.

Corrections in Release 9.1.0.2

The following anomalies are corrected in Release 9.1.0.2.

Anomaly	Description
41403 From the Settings icon on the toolbar, when the user clicks a menu item (such as Configuration), but does not select an option (such as Configuration => Audit Trail), the menu closes	Acuitas now properly handles Application Menu closing when no proper action is associated with the menu item.
41426 Acuitas Historical Statistics do not display instance identifiers for the LTEMME_PagingProfile statistics table	Historical Statistics now display instance identifiers for the LTEMME_PagingProfile statistics table.

Corrections in Release 9.1.0.1

The following anomalies are corrected in Release 9.1.0.1.

Anomaly	Description
40237 For HA, MySQL replication lag can occur for large scale deployment	Relaxed MySQL sync settings to rectify the lagging issue in large scale deployments.

Corrections in Release 9.1.0.0

The following anomalies are corrected in Release 9.1.0.0.

Anomaly	Description
41356 HA-DR - No alarm is raised if either of slave pair is down	emsDBReplication alarm is now correctly raised.
34077/40653 Acuitas receiving the following SNMP alarm: SNMP Agent is down / SNMP Version is not supported	Changed the default that Acuitas uses to determine that a NE is SNMP unreachable. The default is now 4 attempts, each with 15 seconds timeout.
39222 Database backup schedule that runs during the first 30 minutes of server startup can cause resource contention issues	Database backup schedule is no longer run during the first 30 minutes of server startup to avoid resource contention issues
39340 Configuration Navigator - Exception occurs when creating Device Database	Flattened parent-child hierarchy so that attempts to create the following object no longer result in an exception: Service Construct > Device Database
40928 The configureRemoteServerForCsvPush.sh script to push exported performance management files from the local Acuitas server to a remote server fails	The configureRemoteServerForCsvPush.sh script does not push the CSV files to the remote server because the script was using runuser command to execute rsync. Instead of using runuser, the script uses eval to execute the commands in the script.
40661 Multicast is no longer required from HA deployments	Changed Wildfly to use point-to-point instead of multicast in cluster formation. With this change, Acuitas no longer requires multicast for HA setup (configure_ha and server startup).
40911 Acuitas performance data WEEKLY and MONTHLY tables are large and can cause performance issues when inserting/deleting table rows	WEEKLY tables now retain 6 weeks of data instead of 3 months of 3-hour data. MONTHLY tables now retain 1 years of data instead of 3 years of daily data.
41022 Acuitas Standalone DR server does not function correctly when using an interface other than eth0	Acuitas Standalone DR server now functions correctly on the server using non-default IP interface (e.g. eth1).

Anomaly	Description
41129 The following KPIs use incorrect formulas: ■ TCP-Splicing - Server Downlink Throughput (Mbps) ■ TCP-Splicing - Server Uplink Throughput (Mbps)	The KPIs now use the correct formulas: ■ TCP-Splicing - Server Downlink Throughput (Mbps): (TCPSPLICINGSTATSCONNECTIONCHASSIS.num_tcp_splicing_server_connections_bytes_received) * 8 / 1,000,000 / Interval ■ TCP-Splicing - Server Uplink Throughput (Mbps): (TCPSPLICINGSTATSCONNECTIONCHASSIS.num_tcp_splicing_server_connections_bytes_sent) * 8 / 1,000,000 / Interval
41277 The configureRemoteServerForCsvPush.sh script forces you to export Performance Management files to both a remote server and local server	The script now allows you to export locally without also specifying a remote server. The following combinations are now supported: ■ Only Remote Server export ■ Only Local Server export ■ Both Remote and Local Server export

Corrections in Release 9.0.0.0

The following anomalies are corrected in Release 9.0.0.0.

Anomaly	Description
Fixes included in the Acuitas software Red Hat/MySQL Image an-ems-v2.0-rhel-server-7.5-x86_64-mysql-5.7.23	The Acuitas software Red Hat/MySQL Image an-ems-v2.0-rhel-server-7.5-x86_64-mysql-5.7.23 contains the following fixes: 40006 Mysql 5.7.20 memory leak 38261 Acuitas database replication stopped 38557 Acuitas HA + DR NEs in the Global Group are missing from the Network View tab 38997 Acuitas process stopped 39040 Acuitas Performance Manager not working 39077 Acuitas Performance Manager Memory Graph inaccurate
40242 HA server does not start up due to database errors when performing recovery of the derived metrics files	Avoided database connection time out that occurred when performing recovery of derived metrics files. Now, Acuitas retrieves only the last 30 minutes of derived metrics files when performing a recovery.
39485 When forwarding an alarm, Acuitas includes the created time instead of the updated time	Acuitas now includes the updated time in the varbind temsServerTimeStamp in a forwarded alarm.
39447 VAS - Saving Ratio KPI does not export with other KPIs from the same table	When VAS - Saving Ratio KPI is exported with other KPIs that share the same table (for example, VAS - Bytes Saved), the VAS - Saving Ratio is now included in the export.

Anomaly	Description
393430 EMS processes stopped, had to be manually restarted	Acuitas now cleans improper sessions and releases used ports as needed.
39291 OpenStack can leave a VM in an orphaned state when it fails to create it. As a result, Scale In is not enabled for that VM in a not-instantiated state.	VNF Manager now allows for scaling in a VM even when it is not configured in the MCC/MCC MME. As a result, any orphaned VM is removed.
38839 Acuitas uses an unusual data format for device-specific logs (yyyy-MM-dd kk:mm:ss,SSS)	Acuitas now uses the date format of yyyy-MM-dd HH:mm:ss,SSS for device-specific logs.
37077 Configuration Navigator — Unable to set Loopback IP for a GTP Interface on vSSF	On an vSSF, the gtp->interface object is configurable and its fields are correctly populated.
29345/40560 Following an upgrade, the scheduled database trigger time is not preserved	Following an upgrade, the skipped database backup is not made up. In addition, no database backup is performed in the first 30 minutes of server startup.

Corrections in Release 8.2.1.2

The following anomalies are corrected in Release 8.2.1.2.

Anomaly	Description
40242 HA server does not start up due to database errors when performing recovery of the derived metrics files	Avoided database connection time out that occurred when performing recovery of derived metrics files. Now, Acuitas retrieves only the last 30 minutes of derived metrics files when performing a recovery.
38255 Some Acuitas MIB files are malformed	Removed all references to @LAST_UDPATED in the Acuitas MIBs.
39313 Acuitas server statistics report incorrect memory usage values (approaching 100%)	The formula to calculate Acuitas server memory usage is changed to use cached memory as free memory.

Corrections in Release 8.2.1.1

The following anomalies are corrected in Release 8.2.1.1.

Anomaly	Description
39774 Acuitas upgrade to 8.1 failed on the HA cluster	Acuitas HA can now be deployed and started following an upgrade to 8.2.1.1

Corrections in Release 8.2.1.0

The following anomalies are corrected in Release 8.2.1.0.

Anomaly	Description
38067 The hamonitor.log reports that Replication server is down	Increased the Time to Live setting to 128.
39137 Running Outage Report results in the error <code>Request for data timed out</code>	Outage Report now correctly lists the outage data without the timeout error.

Corrections in Release 8.2.0.0

The following anomalies are corrected in Release 8.2.0.0.

Anomaly	Description
36661 PM Dashboard does not always provide datapoint tooltip	When you point the cursor at a data point in the PM Dashboard graph, Acuitas correctly opens a pop-up window to list the node name, timestamp, and data value.
37064 Configuring a Threshold with a high absolute value fails	Changed validation to check the threshold value range from -99999999999.99 to 99999999999.99. Changed the threshold value datatype in the THRESHOLDDetails table to decimal(14,2). When editing the threshold details, the value format is now shown as digits with 2 decimal precision.
37209 Acuitas missing support for configurable Performance Manager Thresholds for SSF counters	For SSF, Acuitas Performance Manager supports configuring Thresholds with various severity levels for a performance statistic attribute or a Key Performance Indicator (KPI). When the data collected for a attribute or KPI crosses the threshold level defined, a threshold event is generated with the specified severity.
37832 Unsupported objects should be hidden in the Configuration Navigator	The following objects are now hidden from the Configuration Navigator as they are not supported by the 8.x MCC: ■ Service Construct > Protocol Application Rule Group ■ Service Construct > Service Rule > Protocol Application Rule Group

Anomaly	Description
38248 Configuration Navigator hangs when attempting to configure vTap > Subscriber Filter > Key	Configuration Navigator no longer hangs when attempting to configure vTap > Subscriber Filter > Key.
38301 Clearing any of the DPI statistics clears all of the DPI statistics tables	Clearing any of the DPI statistics returns a confirmation prompt indicating that the action clears all of the DPI statistics. For example: <pre>zone <name> deep-packet-inspection statistics grid-level general clear</pre> This will clear all of the DPI statistics. Do you wish to proceed? (yes/no) : Any scripts that clear DPI statistics must account for this new prompt.
38317 EMS Server CPU Usage Threshold crossing alarm is not working as expected	Acuitas Performance Manager now correctly handles the Thresholds crossing alarms for EMS Server CPU Usage.
38350 Configuration Navigator missing parameters for IP Protocols > Policy > Access List	Configuration Navigator correctly contains parameters for IP Protocols > Policy > Access List.
38694 Acuitas does not support restoring SSF configuration data across mismatched device versions	Acuitas NE Configuration Backup and Restore now supports restoring SSF configuration data from a lower SSF version backup to a higher device version, such as from SSF 1.3.08 to 2.0.10.

Corrections in Release 8.1.3.0

The following anomalies are corrected in Release 8.1.3.0.

Anomaly	Description
37264, 37972 In the case of a many instances, the real-time support for the following statistics tables could cause a memory crash: ■ EMSGWQCIARPSTATS ■ EMSGWAPNQCARPSTATS	The EMSGWQCIARPSTATS and EMSGWAPNQCARPSTATS are no longer listed in the Real Time Report. These statistics are displayed as Historical statistics only.

Anomaly	Description
37477 Cluster topology alarm is not raised when one of the servers is down	This fix provides the following changes: ■ For Standby Down event, changed the severity level from Warn to Critical. ■ Removed the Standby to Master switchover event. ■ Raise Critical alarm (Standby Down event) when the other node is down. ■ Raise Clear alarm (Standby Up event) when the other node comes up.
37517 configure_ha and configure_dr may run into issues when updating iptables.	configure_ha and configure_dr no longer run into issues when updating iptables.
37884 If an File Server name contains SFTP or TFTP, the file server does not appear in the selection list when configuring software maintenance releases	Acuitas now handles the case of a File Server name containing SFTP or TFTP.
37956 MME Acuitas statistics are abnormal following an MME 12.1.18 and Acuitas 7.3.8 upgrade	Following the upgrade, MME statistics raw files are now processed normally.
38159, 38160 With support to override the following confirmation prompt in the Acuitas Provisioning API and the CLI: <code>This may impact a large number of subscribers. Would you like to proceed? (yes/no)</code> Acuitas needs to enable prompts once the execution completes	Acuitas now re-enables prompts once the script or CLI finishes executing.
38294 Cluster deployment fails when deploying MCC in OpenStack in a non-network-redundant mode.	A cluster without network redundancy is now correctly deployed.
38427 configure_ha fails to set up an HA pair in IPv6 environment.	configure_ha now functions correctly in IPv6 environment.

Corrections in Release 8.1.2.1

The following anomalies are corrected in Release 8.1.2.1.

Critical Anomaly	Description
37454, 37938 In the DNSRESOLUTIONSTATS statistics table, the statistic NumDnsResponseFailOperationNotImplementedByServer is not being collected	In the DNSRESOLUTIONSTATS statistics table, the statistic NumDnsResponseFailOperationNotImplementedByServer is now being correctly collected.

Corrections in Release 8.1.2.0

The following anomalies are corrected in Release 8.1.2.0.

Anomaly	Description
37677 MCC fails if the same file exists in the /home/users/emsadmin directory	When the clear script runs, the system creates a file on the MCC (/home/users/emsadmin#subscriber_output.1). The file is now explicitly removed before attempting a save operation on the MCC. With the fix, the script no longer hangs.
37122 Alarm/Eventlog - purging with a large events database sometimes fails	Alarm/Eventlog - purging no longer fails with a large events database (1 million or more events) associated with a single alarm. Now, purging 1 million entries from the EVENTALARMMAPPING table takes 1-2 minutes (instead of hours or failed).
36923 Acuitas server may become stuck due to database connection leaks	Acuitas server no longer becomes stuck due to database connection leaks.

Corrections in Release 8.1.1.0

The following anomalies are corrected in Release 8.1.1.0.

Anomaly	Description
37439 In some of the DPI sub protocol tables, when a statistics name contains a dash character (-), the historical statistics data is not collected	Now, even when a statistics name contains a dash character (-), the historical statistics data is collected.

Anomaly	Description
37134, 36375 Acuitas security group members with Configuration > SSF Configuration > Read permission should be restricted from making SSF Configuration Navigator changes	Acuitas security group members with Configuration > SSF Configuration > Read permission are restricted from making SSF Configuration Navigator changes.
36120 emsDBReplicationDown alarm indicates the wrong HA server	When emsDBReplicationDown alarm is raised when one of the HA servers is down, the alarm indicates the correct server. For example, for an HA system with server A and server B, if server B is down, the raised alarm correctly states "Replication server (B) is down."
35100 The ems.log file grows over time and is never rotated	The ems.log is now managed by logrotate.d. Acuitas maintains up to 10 logs of 10MB each.
34504 HA - haupgrade console output hangs in the middle	When upgrading an HA cluster, the console output no longer gets stuck after the standby server is upgraded.
31891/23196 The saved CLI configuration file cannot be sourced back to the same MCC/MME and returns an error	A saved CLI configuration file can now be sourced back to the same MCC/MME.

Corrections in Release 8.1.0.0

The following anomalies are corrected in Release 8.1.0.0.

Anomaly	Description
36797 Error from Acuitas scripts when Acuitas is using non-default interface (eth1)	When Acuitas has been setup using a non-default IP interface (eth1), running Acuitas scripts (such as emsstatus, starttems, emsstop, emsha, and emsdb) no longer returns a permission denied error.
36689 User-defined PM Dashboards are not preserved when sourced from a database backup	User-defined PM Dashboards are now preserved when sourced from a database backup.
36645 In the Cluster Configuration screen, the IP address text box is pushed to the next line instead of appearing on the same line as the parameter name.	In the Cluster Configuration screen, the IP address text box now appears on the same line as the parameter name.

Anomaly	Description
36231 Clearing counters in the TCPMEASUREMENTS SERVERSIDE STATS table is not properly supported for MCC 7.2.10 due to the table not having the discontinuity timestamp	Clearing counters in the TCPMEASUREMENTS SERVERSIDE STATS table is now properly supported. The table now contains the discontinuity timestamp.
35495 When an MME state is changed from Managed to Unmanaged, the Configuration Export Manager does not correctly list the nodes of the configuration files	When an MME state is changed from Managed to Unmanaged, the Configuration Export Manager now correctly lists the nodes of the configuration files.

Corrections in Release 8.0.0.0

The following anomalies are corrected in Release 8.0.0.0.

Anomaly	Description
36311 For an MCC configured with advanced-services-separation building block strategy, MCC nodes/VMs without personalities are missing the Scale In/Scale Out Virtual Machine Operations	For an MCC configured with advanced-services-separation building block strategy, MCC nodes/VMs without personalities now have the Scale In/Scale Out Virtual Machine Operations .
35126 Configuration Navigator handles union of type enumeration only	Acuitas now handles union of other defined types and base types.
34561 Query for a large number of alarms causes the server to run out of memory	Set the maximum page size to 1000 to prevent the server from encountering an out of memory error.
34522 Geographic Redundancy group comparison does not identify entities configured for Geographic Redundancy Group	After adding entities to a Geographic Redundancy group (such as Charging Instance, Content Filter Instance, and Data Profile) and then comparing configurations before replicating the group, those entities correctly appear in the comparison.
34445 Alarms filter does not function with alarm forwarding	Applying an alarm filter that contains the Group Contains condition no longer stops all alarms from being forwarded.

Anomaly	Description
34254 Manual scale out fails for a node reserved for expansion	Manual scale out now succeeds for a node Reserved for expansion.
34165 Security Group > Associate Resources - When the Associate All Resources checkbox is marked, all devices should be listed in the Assigned List	From the Security Group > Associate Resources window, when the Associate All Resources checkbox is marked, all NEs are now listed in the Assigned List .
34167 Security Group > Associate Resources - List of NEs does not contain the full list	From the Security Group > Associate Resources window, when the Associate All Resources checkbox is marked, the list of NEs now contains the full list.
33038 Acuitas server hangs when purging a large number of event logs and/or alarms.	Event logs and alarms are purged using smaller chunks to ensure it does not affect overall Acuitas health.
33889 Node 1 assigned as SSM/IOM/WSM leads to an intermittent issue where some session does not process proxy traffic	VNF Manager now prevents Node ID 1 (one) from being assigned to SSM node type of MCC clusters.
32200 Security vulnerabilities	Several major security vulnerabilities are now addressed.
32000 Default MySQL replication configuration does not scale	Configure_dr script will apply respective MySQL replication configuration for master or slave HA so it can utilize the server resources to support larger replication throughput.
31718 Acuitas server becomes inaccessible due to web session leakage	Acuitas 8.0 upgrades Wildfly to 8.2.1 to address potential file descriptor leakages. Also, web session management has been reworked.
28350 Following an upgrade to 8.0.0.0, alarm acknowledge status is not preserved	Following an upgrade to 8.0.0.0, the value of the Ack. Status field is now preserved.

Documentation Updates and Corrections

This section lists any new features, documentation updates, and corrections that were not included in the released documentation set.

Update to Configuration Options for Cluster > Node > Admin State

The **disabled** and **maintenance** options for Cluster > Node > Admin State described in the *Acuitas User's Guide* are not supported and should not be used.

The following use case examples describe how to perform maintenance actions on the cluster, and increase and reduce system capacity.

Performing Maintenance Actions

- 1 Use the hypervisor's VM shutdown procedure to sequentially power off each VM running on the host.
- 2 Wait for redundancy to transfer service to the standby VMs and for the system to stabilize.
- 3 Perform the required maintenance on the host.
- 4 Sequentially power up the VMs, making sure that the system stabilizes after each VM powers up.

Increasing/Decreasing System Capacity

CAUTION: The following procedure impacts service. Either move service off the MCC or perform these actions during periods of decreased traffic.

- 1 Sequentially enable or disable the VMs.
- 2 Reboot the cluster.
- 3 Transfer service back to the MCC.